

CHỨNG MINH ĐỊNH LÝ LỚN FERMAT

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Nhà toán học người Pháp Pierre de Fermat (1601–1665) đã đặt ra một định lý nổi tiếng: “**phương trình $x^n + y^n = z^n$, trong đó n là một số nguyên lớn hơn 2, không thể tìm được nghiệm nguyên thỏa mãn $xyz \neq 0$.**”. Đây là định lý Fermat cuối cùng. Ông đã viết lên lề cuốn sách một tuyên bố nhưng không cung cấp chứng minh, chỉ để lại dòng chú thích: “Tôi có một phương pháp rất xuất sắc để chứng minh cho trường hợp tổng quát, nhưng không thể viết ra được ở đây vì lề sách quá hẹp!”

Các nhà toán học sau đó đã cố gắng giải quyết bài toán này trong suốt gần bốn thế kỷ. Cuối cùng, nhà toán học Andrew Wiles (người Anh, định cư tại Mỹ, sinh năm 1953) đã công bố lời giải duy nhất và không thể nào cản trở được vào mùa hè năm 1993, và sau đó là một phiên bản sửa đổi vào năm 1995.

Trong một khoảng thời gian dài, tôi đã cố gắng chứng minh định lý này bằng cách sử dụng lý thuyết số cơ bản và cuối cùng đã thành công. Dưới đây là toàn văn của phần chứng minh định lý Fermat lớn.

Để chứng minh định lý Fermat lớn này, tôi trình bày bảy bổ đề và một phương pháp biến đổi tổng $\sum$, cùng với việc sử dụng đạo hàm cấp k của hàm số đa thức một biến để tính tổng $\sum$.

A. CÁC BỔ ĐỀ

Bổ đề 1:

Cho $n \in \mathbb{N}, n \geq 3$. Nếu phương trình (PT) $x^n + y^n = z^n$ (1) có nghiệm nguyên khác không thì bao giờ cũng tồn tại nghiệm nguyên $x = u, y = v, z = t$ sao cho $(u, v) = (u, t) = (v, t) = 1$.

Bổ đề 2:

Nếu có $n \in \mathbb{N}, n \geq 3$, PT $x^n + y^n = z^n$ vô nghiệm nguyên khác không thì với mọi $k \in \mathbb{N}^*$, PT $x^{nk} + y^{nk} = z^{nk}$ cũng vô nghiệm nguyên khác không.

Bổ đề 3:

Cho $a.b = c^n$, n là số nguyên dương lẻ; $a, b, c \in \mathbb{Z}^*$ sao cho $(a, b) = 1$, khi đó tồn tại hai số nguyên khác không c_1, c_2 sao cho $(c_1, c_2) = 1, c = c_1.c_2, a = c_1^n$ và $b = c_2^n$.

Bổ đề 4:

Cho $a.b = c^n$, n là số nguyên dương lẻ, $a, b, c, m \in \mathbb{Z}^*$, sao cho $|a| \geq 2, |b| \geq 2, |c| \geq 2, m$ là số nguyên tố, $(a, b) = m, b \nmid m^2$, khi đó tồn tại số nguyên dương s sao cho $ns \geq 2$ và hai số nguyên khác không c_1, c_2 sao cho $(c_1, c_2) = 1$ đồng thời $c_1 \nmid m, c_2 \nmid m, c = m^s.c_1.c_2, a = m^{ns-1}.c_2^n$ và $b = m.c_1^n$.

Bổ đề 5:

Cho n là số nguyên tố lẻ và hai số nguyên u, v sao cho $(u, v) = 1$, khi đó:

a) Nếu $(u + v) \nmid n$ thì $\left(u + v, \sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i\right) = 1$ và $\left(\sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i\right) \nmid n$.

b) Nếu $(u + v) : n$ thì $\left(u + v, \sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i\right) = n$ và $\left(\sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i\right) \nmid n^2$.

c) Nếu $(u^n + v^n) : n$ thì $(u^n + v^n) : n^2$ (không cần giả thiết $(u, v) = 1$).

d) Nếu $u \nmid n$ thì $u^{n(n-1)} \equiv 1 \pmod{n^2}$.

Bổ đề 6:

Cho n là số nguyên tố lẻ. Giả sử PT: $x^n + y^n = z^n$ (1) có nghiệm nguyên khác không $x = u, y = v, z = t$ sao cho $(u, v) = (u, t) = (v, t) = 1$ và $uv \nmid n$. Khi đó tồn tại 4 số nguyên a, b, c, k khác không, đôi một nguyên tố cùng nhau và không chia hết cho n , đồng thời tồn tại số nguyên s lớn hơn 1, sao cho: $v = a^n + n^s.abck, u = b^n + n^s.abck, t = h - n^s.abck, a^n + b^n = h - 2n^s.abck$; với $h = c^n$, khi $t \nmid n$ hoặc $h = n^{ns-1}.c^n$, khi $t : n$.

Bổ đề 7:

Nếu n là số nguyên tố lớn hơn 3 và ba số nguyên a, b, c , thỏa mãn

$abc \not\equiv 0 \pmod{n}$, $a^n + b^n - c^n \equiv 0 \pmod{n^2}$, $a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv 0 \pmod{n}$ thì ta có $(a^n - b^n)(c^n + a^n)(c^n + b^n) \not\equiv 0 \pmod{n}$
 $a^{n^2} + b^{n^2} - c^{n^2} \equiv 0 \pmod{n^2}$, $a^{n(n-3)} + b^{n(n-3)} + c^{n(n-3)} \equiv c^{3n} + b^{3n} \equiv a^{3n} - b^{3n} \equiv b^n c^n + a^{2n} \equiv c^n a^n + b^{2n} \equiv b^n a^n - c^{2n} \equiv 0 \pmod{n^2}$, $n \equiv 1 \pmod{6}$
 $a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv a^{n(n-4)} + c^{n(n-4)} \equiv b^{n(n-4)} + c^{n(n-4)} \equiv a^{n(n-4)} - b^{n(n-4)} \equiv 0 \pmod{n^2}$.

B. MỘT CÁCH BIẾN ĐỔI TỔNG $\sum$ VÀ MỘT DẠNG TÍNH ĐƯỢC CỦA TỔNG $\sum$.

$$1) \sum_{i=k}^n a_i = \sum_{i=m}^{n+m-k} a_{i-m+k}, k, m \in \mathbb{N}$$

$$2) \sum_{i=k}^{m-1} i(i-1)(i-2)\dots(i-k+1)h^{m-1-i}x^{i-k} = \sum_{i=0}^{m-1-k} (m-1-i)(m-2-i)\dots(m-k-i)h^i x^{m-1-k-i} = f^{(k)}(x), k \in \mathbb{N}^*$$

, với $f(x) = x^{m-1} + hx^{m-2} + \dots + h^{m-2}x + h^{m-1} = \frac{x^m - h^m}{x-h}$, $x \neq h$; $f^{(k)}(x)$ là đạo hàm cấp k của hàm số $f(x)$, $m \in \mathbb{N}$, $m \geq 2$.

C. CHỨNG MINH CÁC BỔ ĐỀ

1. Chứng minh bổ đề 1

Giả sử tồn tại số nguyên n lớn hơn 2, sao cho PT(1) có $x = u_0$, $y = v_0$, $z = t_0$ là một nghiệm nguyên khác không và $(u_0, v_0, t_0) = d$. Khi đó tồn tại ba số $u, v, t \in \mathbb{Z}^*$ sao cho $(u, v, t) = 1$, $u_0 = ud$, $v_0 = vd$, $t_0 = td$ và $u_0^n + v_0^n = t_0^n$, suy ra $u^n + v^n = t^n$ (1'), suy ra $x = u$, $y = v$, $z = t$ là một nghiệm nguyên khác không của PT(1).

Giả sử $(u, v) = d'$, từ (1') suy ra $t^n : d'^n$, suy ra $t : d'$, mà $(u, v, t) = 1$ nên $d' = 1$. Vậy $(u, v) = 1$.

Chúng minh tương tự ta cũng có $(u, t) = (t, v) = 1$. **Bổ đề 1** đã được chứng minh.

2. Chứng minh bổ đề 2

Giả sử tồn tại số nguyên dương k_0 sao cho PT $x^{nk_0} + y^{nk_0} = z^{nk_0}$ có nghiệm nguyên khác không $x = u$, $y = v$, $z = t$, khi đó ta có $u^{nk_0} + v^{nk_0} = t^{nk_0} \Leftrightarrow (u^{k_0})^n + (v^{k_0})^n = (t^{k_0})^n$, suy ra PT $x^n + y^n = z^n$ có nghiệm nguyên khác không $x = u^{k_0}$, $y = v^{k_0}$, $z = t^{k_0}$, điều này trái với giả thiết. **Bổ đề 2** đã được chứng minh.

3. Chứng minh bổ đề 3

Giả sử $(a, c) = c'_1$, khi đó tồn tại hai số nguyên a_1, c'_2 sao cho $(a_1, c'_2) = 1$ và $a = c'_1 a_1$, $c = c'_1 c'_2$. Vì $ab = c^n$, nên $c'_1 a_1 b = c_1'^n c_2'^n$, suy ra $a_1 b = c_1'^{n-1} c_2'^n$, suy ra $b | c_1'^{n-1} c_2'^n$. Vì $(c'_1, b) = 1$ (do $(a, b) = 1$) nên $c_2'^n = kb$, với k là số nguyên khác không, từ đẳng thức $a_1 b = c_1'^{n-1} c_2'^n$, ta suy ra $a_1 = k c_1'^{n-1}$, suy ra $a_1^n = k^n c_1'^{n(n-1)}$, suy ra $k | a_1^n$ và $k | c_2'^n$; vì $(a_1, c'_2) = 1$ nên $(a_1^n, c_2'^n) = 1$, từ đó suy ra $|k| = 1$.

Vậy $b = k(kb) = k c_2'^n = (k c'_2)^n = c_2^n$, $a = k(ka) = k c_1'^n = (k c'_1)^n = c_1^n$, $c = c_1 c_2$, (vì $|k| = 1$, n lẻ) với $c_1 = k c'_1$, $c_2 = k c'_2$ nguyên tố cùng nhau.

Bổ đề 3 đã được chứng minh.

4. Chứng minh bổ đề 4

Vì $a : m$, $b : m$ và $ab = c^n$, suy ra $c^n : m^2$, suy ra $c : m$ (do m là số nguyên tố). Giả sử $c = m^s r$, với $s \in \mathbb{N}^*$, $r \in \mathbb{Z}^*$ và $r \not\equiv m$, mà $c^n = m^{ns} r^n : m^2$, nên $ns \geq 2$. Do $(a, b) = m$ và $b \not\equiv m^2$, nên có thể giả sử $b = mh$ và $a = m^k l$, với $k \in \mathbb{N}^*$; $h, l \in \mathbb{Z}^*$ sao cho $(h, l) = 1$, $h \not\equiv m$, $l \not\equiv m$. Vì $ab = c^n$ nên $m^{k+1} hl = m^{ns} r^n$, suy ra $k+1 = ns$, suy ra $k = ns - 1$ và $hl = r^n$; áp dụng **bổ đề 3**, ta có $r = c_1 c_2$, $h = c_1^n$, $l = c_2^n$, trong đó hai số c_1, c_2 là hai số nguyên sao cho $(c_1, c_2) = 1$. Từ đó ta có $c = m^s c_1 c_2$, $a = m^{ns-1} c_2^n$, $b = m c_1^n$.

Bổ đề 4 đã được chứng minh.

5. Chứng minh bổ đề 5

a) Đặt $A = \sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i$. Ta có $u^n + v^n = [(u+v) - v]^n + v^n = (u+v) \sum_{k=0}^{n-1} (-1)^k C_n^k (u+v)^{n-1-k} v^k$, $u^n + v^n = (u+v)A$ mà $u+v \neq 0$, nên $A = \sum_{k=0}^{n-1} (-1)^k C_n^k (u+v)^{n-k-1} v^k$. Ta có $A = (u+v) \sum_{k=0}^{n-2} (-1)^k C_n^k (u+v)^{n-k-2} v^k + nv^{n-1}$, $(u, v) = 1$ và $(u+v) \not\equiv n$

nên $(u+v, nv^{n-1})=1$, suy ra $(A, u+v)=1$ (đpcm).

b) Ta có $A = n(u+v) \left[\sum_{k=0}^{n-3} (-1)^k \frac{C_n^k}{n} (u+v)^{n-k-2} v^k - \frac{(n-1)v^{n-2}}{2} \right] + nv^{n-1} = n(u+v)A_1 + nv^{n-1}$, với $\sum_{k=0}^{n-3} (-1)^k \frac{C_n^k}{n} (u+v)^{n-k-2} v^k - \frac{(n-1)v^{n-2}}{2} = A_1$. Với $k = \overline{1, n-1}$ thì $C_n^k : n$ (vì n là số nguyên tố), mà $(u+v) : n$ nên $n(u+v)A_1 : n^2$; mặt khác $nv^{n-1} : n$ và $nv^{n-1} \not\vdash n^2$ (do $v \not\vdash n$), từ đó suy ra $n(u+v)A_1 + nv^{n-1} = A : n$ và $A \not\vdash n^2$, hơn nữa $(u+v, A) = n$ (đpcm).

c) * Nếu $uv : n$ thì $u : n \vee v : n$ (vì n là số nguyên tố), suy ra $u^n : n \vee v^n : n$ và do $(u^n + v^n) : n$ nên $u^n : n \wedge v^n : n$, suy ra $u : n \wedge v : n$ (vì n là số nguyên tố), suy ra $(u^n + v^n) : n^2$ (đpcm).

* Nếu $uv \not\vdash n$ thì $u \not\vdash n, v \not\vdash n$, suy ra $u^{n-1} \equiv v^{n-1} \equiv 1 \pmod{n}$ (định lý nhỏ Fermat) suy ra $u^{n-1} - v^{n-1} = (u^{n-1} - 1) - (v^{n-1} - 1) \equiv 0 \pmod{n}$.

Giả sử $(u+v) \not\vdash n$, ta có $\sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i = u^{n-1} - v \sum_{i=0}^{n-2} (-1)^i u^{n-2-i} v^i = u^{n-1} - \frac{v(u^{n-1} - v^{n-1})}{u+v} \equiv u^{n-1} \equiv 1 \pmod{n}$, suy ra $\sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i \not\vdash n$, suy ra $u^n + v^n = (u+v) \sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i \not\vdash n$, trái với giả thiết.

Vậy $(u+v) : n$, theo một kết quả chứng minh ở b) là $\sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i = A : n$, suy ra $u^n + v^n = (u+v) \sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i : n^2$ (đpcm).

d) Vì n là số nguyên tố và $u \not\vdash n$ nên $u^{n-1} \equiv 1 \pmod{n}$ (định lý nhỏ Fermat), suy ra $u^{n-1} = 1 + mn, m \in \mathbb{N}$.
 $\Rightarrow u^{n(n-1)} = (u^{n-1})^n = (1 + mn)^n \equiv 1 \pmod{n^2}$ (đpcm).

Bổ đề 5 đã được chứng minh.

6. Chứng minh bổ đề 6.

Vì $x=u, y=v, z=t$ là nghiệm của PT (1) nên ta có $u^n + v^n = t^n$ (3), suy ra: $(u+v) \sum_{i=0}^{n-1} (-1)^i u^{n-i-1} v^i = t^n$ (4),

$$(t-v) \sum_{i=0}^{n-1} t^{n-i-1} v^i = u^n \quad (5), \quad (t-u) \sum_{i=0}^{n-1} t^{n-i-1} u^i = v^n \quad (6)$$

Xét hai trường hợp của t .

6.1. Khi t không chia hết cho n .

Ta có $(u, v) = (u, t) = (v, t) = 1$ và cùng với (4), (5), (6) suy ra $(u+v) \not\vdash n, (t-v) \not\vdash n$ và $(t-u) \not\vdash n$; áp dụng **bổ đề 5** ta có $\left(u+v, \sum_{i=0}^{n-1} (-1)^i u^{n-i-1} v^i \right) = \left(t-v, \sum_{i=0}^{n-1} t^{n-i-1} v^i \right) = \left(t-u, \sum_{i=0}^{n-1} t^{n-i-1} u^i \right) = 1$, $\left(\sum_{i=0}^{n-1} (-1)^i u^{n-i-1} v^i \right) \not\vdash n, \left(\sum_{i=0}^{n-1} t^{n-i-1} v^i \right) \not\vdash n, \left(\sum_{i=0}^{n-1} t^{n-i-1} u^i \right) \not\vdash n$, kết hợp với (4), (5), (6) và áp dụng **bổ đề 3**, ta có: $u+v = c^n$ (4') và $\sum_{i=0}^{n-1} (-1)^i u^{n-i-1} v^i = c'^n$, với

$$t = c.c', c, c' \in \mathbb{Z}^*, (c, c') = 1 \text{ và } c \not\vdash n, c' \not\vdash n; t-v = b^n \quad (5') \text{ và } \sum_{i=0}^{n-1} t^{n-i-1} v^i = b'^n, \text{ với}$$

$$u = b.b', b, b' \in \mathbb{Z}^*, (b, b') = 1 \text{ và } b \not\vdash n, b' \not\vdash n; t-u = a^n \quad (6') \text{ và } \sum_{i=0}^{n-1} t^{n-i-1} u^i = a'^n, \text{ với } v = a.a', a, a' \in \mathbb{Z}^*, (a, a') = 1,$$

$$\text{và } a \not\vdash n, a' \not\vdash n; (a, b) = (a, c) = (c, b) = 1 ((a, b) = (a, c) = (c, b) = 1 \text{ (vì } (b.b', a.a') = (b.b', c.c') = (a.a', c.c') = 1))$$

Ta có: $c'^n - 1 = \sum_{k=1}^{n-1} (-1)^k C_n^k (u+v)^{n-k-1} v^k + [(u+v)^{n-1} - 1]$. Vì n là số nguyên tố và $(u+v) \not\vdash n$, nên với $k = \overline{1, n-1}$

thì $C_n^k : n$ và $[(u+v)^{n-1} - 1] : n$ (định lý nhỏ Fermat). Do đó $(c^n - 1) : n$, mà $c' \nmid n$ nên $(c'^{n-1} - 1) : n$, suy ra $c'^n - 1 - (c'^{n-1} - 1) = c'^{n-1}(c' - 1) : n$, suy ra $(c' - 1) : n$, suy ra $c' = 1 + nk_3$, với $k_3 \in \mathbb{Z}$. Chứng minh tương tự ta cũng có $a' = 1 + nk_1$, $b' = 1 + nk_2$, với $k_1, k_2 \in \mathbb{Z}$, suy ra $t^n - u - v = (c.c')^n - c^n = c^n [(1 + nk_3)^n - 1] : n^2$. Chứng minh tương tự ta cũng có $(u^n - t + v) : n^2$ và $(v^n - t + u) : n^2$ suy ra $(u^n - t + v) + (v^n - t + u) - (t^n - u - v) = 2(u + v - t) : n^2$, suy ra $(u + v - t) : n^2$. Từ (4'), (5'), (6') và $(a, b) = (a, c) = (c, b) = 1$ ta có $u + v - t = c(c^{n-1} - c') = a(a' - a^{n-1}) = b(b' - b^{n-1}) = n^s abck$, với $k \in \mathbb{Z}, s \in \mathbb{Z}, s \geq 2, (k, a) = (k, b) = (k, c) = 1, k \nmid n$, nếu $k \neq 0$. Suy ra $u = b^n + n^s abck, v = a^n + n^s abck, t = c^n - n^s abck, a^n + b^n = c^n - 2n^s abck$.

6.2. Khi t chia hết cho n

Từ (4) suy ra $(u + v) : n$, áp dụng **bổ đề 5b**) ta có $\left(u + v, \sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i\right) = n$ và $\left(\sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i\right) \nmid n^2$, kết hợp với (4) và áp dụng **bổ đề 4**, ta có: $u + v = n^{sn-1} c^n$ (4'') và $\sum_{i=0}^{n-1} (-1)^i u^{n-1-i} v^i = nc'^n$ với $t = n^s c.c', s \in \mathbb{N}^*; c, c' \in \mathbb{Z}^*, (c, c') = 1, c \nmid n, c' \nmid n$. Ta có $t^n - u - v = (n^s c.c')^n - n^{sn-1} c^n = n^{sn-1} c^n [nc'^n - 1] : n^2$ (5'') (vì n là số nguyên tố lẻ, $s \in \mathbb{N}^*$) Từ (5), (6) suy ra $(t - v) \nmid n$ và $(t - u) \nmid n$, áp dụng **bổ đề 5a**) ta có $\left(t - v, \sum_{i=0}^{n-1} t^{n-i-1} v^i\right) = \left(t - u, \sum_{i=0}^{n-1} t^{n-i-1} u^i\right) = 1$, $\left(\sum_{i=0}^{n-1} t^{n-i-1} v^i\right) \nmid n, \left(\sum_{i=0}^{n-1} t^{n-i-1} u^i\right) \nmid n$, kết hợp với (5), (6) và áp dụng **bổ đề 3** ta có $t - v = b^n$ (5') và $\sum_{i=0}^{n-1} t^{n-i-1} v^i = b^n$, với $u = b.b', b, b' \in \mathbb{Z}^*, (b, b') = 1, b \nmid n, b' \nmid n; t - u = a^n$ (6'), với $v = a.a', a, a' \in \mathbb{Z}^*, (a, a') = 1, a \nmid n, a' \nmid n; (a, b) = (a, c) = (c, b) = 1$.

Lập luận như **6.1.** ta có $b' = 1 + nk_2, a' = 1 + nk_1$, suy ra $u^n - t + v = (b.b')^n - b^n = b^n [(1 + nk_2)^n - 1] : n^2$ (5'') (và $v^n - t + u = (a.a')^n - a^n = a^n [(1 + nk_1)^n - 1] : n^2$ (6''), kết hợp (4''), (5'') và (6'') suy ra $(u^n - t + v) + (v^n - t + u) - (t^n - u - v) = 2(u + v - t) : n^2$, suy ra $(u + v - t) : n^2$, mà $u + v - t = n^s c(n^{ns-1-s} c^{n-1} - c') = a(a' - a^{n-1}) = b(b' - b^{n-1}) = n^s abck$ nên $s \geq 2; (k, a) = (k, b) = (k, c) = 1$ và $k \nmid n$, nếu $k \neq 0$. Suy ra $v = a^n + n^s abck, u = b^n + n^s abck, t = n^{ns-1} c^n - n^s abck, a^n + b^n = n^{ns-1} c^n - 2n^s abck$.

6.2.1. Khi k = 0

Ta nhận thấy khi $k = 0$ thì $t - u = v, t - v = u$, hay $a^n = a.a', b^n = b.b',$ suy ra $a^{n-1} = a', b^{n-1} = b' \Rightarrow |a| = |a'| = |b| = |b'| = 1$ (vì $(a, a') = (b, b') = 1$)
 $\Rightarrow t = 0 \vee |t|^n = 2$, vô lý.

Vậy, trong trường hợp $k = 0$, PT (1) vô nghiệm nguyên khác không.

Theo **6.1.** và **6.2.**, ta có **kết quả quan trọng**: $v = a^n + n^s abck, u = b^n + n^s abck, t = h - n^s abck, a^n + b^n = h - 2n^s abck$; với $h = c^n$, khi $t \nmid n$ hoặc $h = n^{ns-1} c^n$, khi $t : n$, trong đó a, b, c, k là bốn số nguyên khác không và đôi một nguyên tố cùng nhau, đồng thời không chia hết cho $n, s \in \mathbb{Z}, s \geq 2$ (đpcm).

7. Chứng minh bổ đề 7.

* Vì $abc \nmid n$, nên $a \nmid n, b \nmid n, c \nmid n$, vì vậy theo **bổ đề 5d**), ta có: $c^{n(n-1)} \equiv a^{n(n-1)} \equiv b^{n(n-1)} \equiv 1 \pmod{n^2}$, suy ra $a^{n^2} + b^{n^2} - c^{n^2} = (a^n + b^n - c^n) - c^n [c^{n(n-1)} - 1] + a^n [a^{n(n-1)} - 1] + b^n [b^{n(n-1)} - 1] \equiv 0 \pmod{n^2}$
 $\Rightarrow a^{n^2} + b^{n^2} - c^{n^2} \equiv 0 \pmod{n^2}$ (a).

* Ta có $c^n \equiv a^n + b^n \pmod{n^2}$ (b) và $c^{n(n-2)} \equiv a^{n(n-2)} + b^{n(n-2)} \pmod{n}$ (c). Từ (c) suy ra $c^{n(n-1)} a^n b^n \equiv a^{n(n-1)} c^n b^n + b^{n(n-1)} c^n a^n \pmod{n}$ (nhân hai vế của đồng dư thức (c) cho $a^n b^n c^n \not\equiv 0 \pmod{n}$)
 $\Rightarrow a^n b^n \equiv c^n (a^n + b^n) \pmod{n}$

$$\Rightarrow a^n b^n \equiv c^{2n} \pmod{n} \text{ (d) (vì } a^n + b^n \equiv c^n \pmod{n^2} \text{)}$$

$$\Rightarrow a^n b^n \equiv (a^n + b^n)^2 \pmod{n}$$

$$\Rightarrow b^{2n} + a^n b^n + a^{2n} \equiv 0 \pmod{n}$$

$$\Rightarrow b^{2n} + a^n (a^n + b^n) \equiv 0 \pmod{n}$$

$$\Rightarrow b^{2n} + a^n c^n \equiv 0 \pmod{n} \text{ (vì } a^n + b^n \equiv c^n \pmod{n^2} \text{)}$$

$$\Rightarrow b^{3n} + c^n (a^n b^n) \equiv 0 \pmod{n}$$

$$\Rightarrow b^{3n} + c^{3n} \equiv 0 \pmod{n} \text{ (vì } a^n b^n \equiv c^{2n} \pmod{n} \text{)}$$

$$\Rightarrow (b^{3n})^n + (c^{3n})^n \equiv 0 \pmod{n^2} \text{ (áp dụng } \textbf{bổ đề 5c})$$

$$\Rightarrow (b^3)^{n(n-1)} \cdot b^{3n} + (c^3)^{n(n-1)} \cdot c^{3n} \equiv 0 \pmod{n^2}$$

$$\Rightarrow b^{3n} + c^{3n} \equiv 0 \pmod{n^2} \text{ (áp dụng } \textbf{bổ đề 5d): } (b^3)^{n(n-1)} \equiv (c^3)^{n(n-1)} \equiv 1 \pmod{n^2} \text{)}$$

Lưu ý rằng trong hai giả thiết đồng dư (c) và (b) của **Bổ đề 7** có mỗi cặp $(a; b), (c; -a), (c; -b)$ đối xứng nhau, nó bình đẳng và có cùng cấu trúc số học trong vành $\mathbb{Z}$. Cho nên, nếu có $b^{3n} + c^{3n} \equiv 0 \pmod{n^2}$ thì ta cũng có $a^{3n} + c^{3n} \equiv a^{3n} + (-b)^{3n} \equiv 0 \pmod{n^2}$ $a^{3n} + c^{3n} \equiv a^{3n} - b^{3n} \equiv 0 \pmod{n^2}$. hay $a^{3n} + c^{3n} \equiv a^{3n} - b^{3n} \equiv 0 \pmod{n^2}$.

* Nếu $b^n + c^n \equiv 0 \pmod{n}$ thì $c^n \equiv -b^n \pmod{n}$, kết hợp với (b) và (c), ta có $a^n \equiv -2b^n \pmod{n}$ và $(-b^n)^{n-2} \equiv a^{n(n-2)} + b^{n(n-2)} \pmod{n}$

$$\Rightarrow a^{n(n-2)} + 2b^{n(n-2)} \equiv 0 \pmod{n}, \text{ (vì } n \text{ lẻ)}$$

$$\Rightarrow (-2b^n)^{n-2} + 2b^{n(n-2)} \equiv 0 \pmod{n}$$

$$\Rightarrow -2^{n-2} b^{n(n-2)} + 2b^{n(n-2)} \equiv 0 \pmod{n}$$

$$\Rightarrow -2^{n-2} + 2 \equiv 0 \pmod{n} \text{ (vì } b \not\equiv n \text{)}$$

$$\Rightarrow -(2^{n-1} - 1) + 3 \equiv 0 \pmod{n}, \text{ (vì } 2^{n-1} - 1 \equiv 0 \pmod{n} \text{), định lý nhỏ Fermat)}$$

$$\Rightarrow 3 \equiv 0 \pmod{n}$$

$$\Rightarrow n = 3, \text{ vô lý.}$$

Vậy $b^n + c^n \not\equiv 0 \pmod{n}$. Chứng minh tương tự ta cũng có $a^n + b^n \not\equiv 0 \pmod{n}, c^n + a^n \not\equiv 0 \pmod{n}$. Tóm lại, ta luôn có: $(a^n - b^n)(c^n + a^n)(c^n + b^n) \not\equiv 0 \pmod{n}$ (đpcm).

* Vì $c^n + b^n \not\equiv 0 \pmod{n}, b^{3n} + c^{3n} = (b^n + c^n)[(c^n - b^n)^2 + b^n c^n] \equiv (b^n + c^n)(a^{2n} + b^n c^n) \equiv 0 \pmod{n^2}$ nên $a^{2n} + b^n c^n \equiv 0 \pmod{n^2}$

Chứng minh tương tự ta cũng có $b^{2n} + a^n c^n \equiv c^{2n} - a^n b^n \equiv 0 \pmod{n^2}$.

* Ta có $c^{2n} - a^n b^n \equiv 0 \pmod{n^2}$

$$\Rightarrow c^n (a^n + b^n) - a^n b^n \equiv 0 \pmod{n^2} \text{ (vì } a^n + b^n \equiv c^n \pmod{n^2} \text{)}$$

$$\Rightarrow c^n a^n \cdot 1 + c^n b^n \cdot 1 - a^n b^n \cdot 1 \equiv 0 \pmod{n^2}$$

$$\Rightarrow c^n a^n b^{n(n-1)} + c^n b^n a^{n(n-1)} - a^n b^n c^{n(n-1)} \equiv 0 \pmod{n^2} \text{ (theo } \textbf{bổ đề 5 d}), \text{ ta có } a^{n(n-1)} \equiv b^{n(n-1)} \equiv c^{n(n-1)} \equiv 1 \pmod{n^2} \text{)}$$

$$\Rightarrow b^{n(n-2)} + a^{n(n-2)} - c^{n(n-2)} \equiv 0 \pmod{n^2} \text{ (đpcm).}$$

* Vì n là số nguyên tố lớn hơn 3 nên $n \equiv \pm 1 \pmod{6}$

Giả sử $n \equiv -1 \pmod{6}$, suy ra $n = 6l - 1, l \in \mathbb{N}^*$. Khi đó, từ $a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv 0 \pmod{n^2}$, ta suy ra $a^{3n(2l-1)} + b^{3n(2l-1)} - c^{3n(2l-1)} \equiv 0 \pmod{n}$

$$\Rightarrow a^{3n(2l-1)} + a^{3n(2l-1)} + a^{3n(2l-1)} \equiv 0 \pmod{n^2} \text{ (vì } c^{3n} \equiv -a^{3n} \pmod{n^2}, b^{3n} \equiv a^{3n} \pmod{n^2} \text{)}$$

$$\Rightarrow 3a^{n(n-2)} \equiv 0 \pmod{n^2}, \text{ vô lý (vì } n \text{ là số nguyên tố lớn hơn 3 và } a^{n(n-2)} \not\equiv 0 \pmod{n} \text{).}$$

Vậy $n \equiv 1 \pmod{6}$ (đpcm).

* Vì $n \equiv 1 \pmod{6}$, nên $n = 6l + 1, l \in \mathbb{N}^*$. Do đó
 $a^{n(n-4)} + c^{n(n-4)} = a^{3n(2p-1)} + (c^{3n})^{(2p-1)} \equiv a^{3n(2p-1)} - a^{3n(2p-1)} \equiv 0 \pmod{n} \text{ (vì } c^{3n} \equiv -a^{3n} \pmod{n^2} \text{)},$ suy ra
 $a^{n(n-4)} + c^{n(n-4)} \equiv 0 \pmod{n^2}$.

Chứng minh tương tự ta cũng có: $b^{n(n-4)} + c^{n(n-4)} \equiv 0 \pmod{n^2}$ suy ra $a^{n(n-4)} - b^{n(n-4)} \equiv 0 \pmod{n}$ (đpcm).

D. CHỨNG MINH ĐỊNH LÝ LỚN FERMAT

1. Khi n là số nguyên tố lớn hơn 11

Giả sử có số nguyên tố n lớn hơn 11, sao cho PT (1) có nghiệm nguyên khác không $x=u, y=v, z=t$, trong đó $(u,v)=(u,t)=(v,t)=1$ (áp dụng **bổ đề 1**). Khi đó $u^n + v^n = t^n$ (3). Vì n lẻ và u, v, t là các số nguyên, nên $u^n + v^n = t^n \Leftrightarrow v^n + u^n = t^n \Leftrightarrow (-t)^n + v^n = (-u)^n \Leftrightarrow u^n + (-t)^n = (-v)^n$; do đó, tính chất số học và vai trò của u, v, t là như nhau; cho nên, không mất tính tổng quát ta giả sử $u \nmid n, v \nmid n$, áp dụng **bổ đề 6**, ta có: $v = a^n + n^s abck, u = b^n + n^s abck, t = h - n^s abck, a^n + b^n = h - 2n^s abck$; với $(h = c^n, \text{ khi } t \nmid n) \vee (h = n^{ns-1}c^n, \text{ khi } t \mid n)$, trong đó a, b, c, k là bốn số nguyên khác không và đôi một nguyên tố cùng nhau, đồng thời không chia hết cho n, $s \in \mathbb{Z}, s \geq 2$.

Ta có: (3) $\Rightarrow (a^n + n^s abck)^n + (b^n + n^s abck)^n = (h - n^s abck)^n$

$$\Leftrightarrow h^n - a^{n^2} - b^{n^2} - n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]n^s abck - \frac{n(n-1)}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^2 - \\ - \frac{n(n-1)(n-2)}{6}[a^{n(n-3)} + b^{n(n-3)} + h^{n-3}](n^s abck)^3 - \frac{n(n-1)(n-2)(n-3)}{24}[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4 - \\ - \dots - n(a^n + b^n - h)(n^s abck)^{n-1} - 3(n^s abck)^n = 0, \text{ suy ra}$$

$$* h^n - a^{n^2} - b^{n^2} \equiv n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]n^s abck \pmod{n^{2s+1}} \quad (7)$$

$$* h^n - a^{n^2} - b^{n^2} \equiv n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]n^s abck + \frac{n(n-1)}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^2 \pmod{n^{3s+1}} \quad (8)$$

$$* h^n - a^{n^2} - b^{n^2} \equiv n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]n^s abck + \frac{n(n-1)}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^2 + \\ + \frac{n(n-1)(n-2)}{6}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^3 \pmod{n^{4s+1}} \quad (9), \quad h^n \equiv a^{n^2} + b^{n^2} \pmod{n^{s+1}} \quad (7')$$

$$* h^n - a^{n^2} - b^{n^2} - n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]n^s abck \equiv \frac{n(n-1)}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^2 + \\ + \frac{n(n-1)(n-2)}{6}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^3 + \frac{n(n-1)(n-2)(n-3)}{24}[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4 \pmod{n^{5s+1}} \quad (10)$$

$$\text{Mặt khác ta có } t^n - u^n = v^n \Leftrightarrow (t-u) \sum_{i=0}^{n-1} t^{n-1-i} u^i = v^n \Leftrightarrow a^n \sum_{i=0}^{n-1} (h - n^s abck)^{n-1-i} (b^n + n^s abck)^i = (a^n + n^s abck)^n$$

$$\Leftrightarrow \sum_{i=0}^{n-1} (h - n^s abck)^{n-1-i} (b^n + n^s abck)^i = (a^{n-1} + n^s bck)^n$$

$$\Leftrightarrow \sum_{i=0}^{n-1} \sum_{j=0}^{n-1-i} \sum_{l=0}^i (-1)^j C_{n-1-i}^j C_i^l h^{n-1-i-j} b^{n(i-l)} (n^s abck)^{l+j} = \sum_{i=0}^n C_n^i a^{(n-1)(n-i)} (n^s bck)^i$$

$$\Leftrightarrow \sum_{i=0}^{n-1} \sum_{j=0}^{n-1-i} \sum_{l=0}^i (-1)^j C_{n-1-i}^j C_i^l h^{n-1-i-j} b^{n(i-l)} (n^s abck)^{l+j} = \sum_{i=0}^n C_n^i a^{n(n-1-i)} (n^s abck)^i$$

$$\Leftrightarrow \sum_{i=0}^{n-1} \sum_{j=0}^{n-1-i} \sum_{l=0}^i (-1)^j C_{n-1-i}^j C_i^l h^{n-1-i-j} b^{n(i-l)} (n^s abck)^{l+j} + \sum_{i=4}^{n-1} (-1)^0 C_{n-1-i}^0 C_i^4 h^{n-1-i} b^{n(i-4)} (n^s abck)^4 + \sum_{i=0}^{n-5} (-1)^4 C_{n-1-i}^4 C_i^0 h^{n-5-i} b^{ni} (n^s abck)^4 +$$

$$+ \sum_{i=1}^{n-4} (-1)^3 C_{n-1-i}^3 C_i^1 h^{n-4-i} b^{n(i-1)} (n^s abck)^4 + \sum_{i=3}^{n-2} (-1)^1 C_{n-1-i}^1 C_i^3 h^{n-2-i} b^{n(i-3)} (n^s abck)^4 + \sum_{i=2}^{n-3} (-1)^2 C_{n-1-i}^2 C_i^2 h^{n-3-i} b^{n(i-2)} (n^s abck)^4 +$$

$$+ \sum_{i=1}^{n-3} (-1)^2 C_{n-1-i}^2 C_i^1 h^{n-3-i} b^{n(i-1)} (n^s abck)^3 + \sum_{i=2}^{n-2} (-1)^1 C_{n-1-i}^1 C_i^2 h^{n-2-i} b^{n(i-2)} (n^s abck)^3 +$$

$$+ \sum_{i=0}^{n-4} (-1)^3 C_{n-1-i}^3 C_i^0 h^{n-4-i} b^{ni} (n^s abck)^3 + \sum_{i=3}^{n-1} (-1)^0 C_{n-1-i}^0 C_i^3 h^{n-1-i} b^{n(i-3)} (n^s abck)^3 + \sum_{i=2}^{n-1} (-1)^0 C_{n-1-i}^0 C_i^2 b^{n(i-2)} h^{n-1-i} (n^s abck)^2 +$$

$$\begin{aligned}
& + \sum_{i=0}^{n-3} (-1)^2 C_{n-1-i}^2 C_i^0 b^{ni} h^{n-3-i} (n^s abck)^2 + \sum_{i=1}^{n-2} (-1)^1 C_{n-1-i}^1 C_i^1 b^{n(i-1)} h^{n-2-i} (n^s abck)^2 + \sum_{i=1}^{n-1} (-1)^0 C_{n-1-i}^0 C_i^1 b^{n(i-1)} h^{n-1-i} n^s abck + \\
& + \sum_{i=0}^{n-2} (-1)^1 C_{n-1-i}^1 C_i^0 b^{ni} h^{n-2-i} n^s abck + \sum_{i=0}^{n-1} (-1)^0 C_{n-1-i}^0 C_i^0 h^{n-1-i} b^{ni} \\
& = \sum_{i=5}^n C_n^i a^{(n-1)(n-i)} (n^s bck)^i + a^{n(n-1)} + na^{n(n-2)} n^s abck + \frac{n(n-1)}{2} a^{n(n-3)} (n^s abck)^2 + \frac{n(n-1)(n-2)}{6} a^{n(n-4)} (n^s abck)^3 + \\
& + \frac{n(n-1)(n-2)(n-3)}{24} a^{n(n-5)} (n^s abck)^4 \\
& \Leftrightarrow \sum_{i=0}^{n-1} \sum_{j=0}^{n-1-i} \sum_{l=0}^i (-1)^j C_{n-1-i}^j C_i^l h^{n-1-i-j} b^{n(i-l)} (n^s abck)^{l+j} + \frac{1}{24} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3) h^{n-1-i} b^{n(i-4)} (n^s abck)^4 + \\
& + \frac{1}{24} \sum_{i=0}^{n-5} (n-1-i)(n-2-i)(n-3-i)(n-4-i) h^{n-5-i} b^{ni} (n^s abck)^4 - \frac{1}{6} \sum_{i=1}^{n-4} i(n-1-i)(n-2-i)(n-3-i) h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \\
& - \frac{1}{6} \sum_{i=3}^{n-2} (n-1-i)i(i-1)(i-2) h^{n-2-i} b^{n(i-3)} (n^s abck)^4 + \frac{1}{4} \sum_{i=2}^{n-3} (n-1-i)(n-2-i)i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 + \\
& + \sum_{i=1}^{n-3} i(n-1-i)(n-2-i) h^{n-3-i} b^{n(i-1)} (n^s abck)^3 - \frac{1}{2} \sum_{i=2}^{n-2} i(i-1)(n-1-i) h^{n-2-i} b^{n(i-2)} (n^s abck)^3 - \\
& - \frac{1}{6} \sum_{i=0}^{n-4} (n-1-i)(n-2-i)(n-3-i) h^{n-4-i} b^{ni} (n^s abck)^3 + \frac{1}{6} \sum_{i=3}^{n-1} i(i-1)(i-2) h^{n-1-i} b^{n(i-3)} (n^s abck)^3 + \\
& + \frac{1}{2} \sum_{i=2}^{n-1} i(i-1) b^{n(i-2)} h^{n-1-i} (n^s abck)^2 + \frac{1}{2} \sum_{i=0}^{n-3} (n-1-i)(n-2-i) b^{ni} h^{n-3-i} (n^s abck)^2 - \sum_{i=1}^{n-2} i(n-1-i) b^{n(i-1)} h^{n-2-i} (n^s abck)^2 + \\
& + \sum_{i=1}^{n-1} i b^{n(i-1)} h^{n-1-i} n^s abck - \sum_{i=0}^{n-2} (n-1-i) b^{ni} h^{n-2-i} n^s abck + \sum_{i=0}^{n-1} h^{n-1-i} b^{ni} \\
& = \sum_{i=5}^n C_n^i a^{(n-1)(n-i)} (n^s bck)^i + a^{n(n-1)} + na^{n(n-2)} n^s abck + \frac{n(n-1)}{2} a^{n(n-3)} (n^s abck)^2 + \frac{n(n-1)(n-2)}{6} a^{n(n-4)} (n^s abck)^3 + \\
& + \frac{n(n-1)(n-2)(n-3)}{24} a^{n(n-5)} (n^s abck)^4 \\
& \Rightarrow \frac{1}{24} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3) h^{n-1-i} b^{n(i-4)} (n^s abck)^4 + \frac{1}{24} \sum_{i=0}^{n-5} (n-1-i)(n-2-i)(n-3-i)(n-4-i) h^{n-5-i} b^{ni} (n^s abck)^4 + \\
& + \frac{1}{6} \sum_{i=1}^{n-4} i(1+i)(2+i)(3+i) h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \frac{n}{6} \sum_{i=1}^{n-4} i(3i^2 + 12i + 11) h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \\
& - \frac{n}{6} \sum_{i=3}^{n-2} i(i-1)(i-2) h^{n-2-i} b^{n(i-3)} (n^s abck)^4 + \frac{1}{6} \sum_{i=3}^{n-2} (1+i)i(i-1)(i-2) h^{n-2-i} b^{n(i-3)} (n^s abck)^4 - \\
& - \frac{n}{4} \sum_{i=2}^{n-3} (3+2i)i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 + \frac{1}{4} \sum_{i=2}^{n-3} (2+i)(1+i)i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 + \\
& + \sum_{i=1}^{n-3} i(n-1-i)(n-2-i) h^{n-3-i} b^{n(i-1)} (n^s abck)^3 - \frac{1}{2} \sum_{i=2}^{n-2} i(i-1)(n-1-i) h^{n-2-i} b^{n(i-2)} (n^s abck)^3 - \\
& - \frac{1}{6} \sum_{i=0}^{n-4} (n-1-i)(n-2-i)(n-3-i) h^{n-4-i} b^{ni} (n^s abck)^3 + \frac{1}{6} \sum_{i=3}^{n-1} i(i-1)(i-2) h^{n-1-i} b^{n(i-3)} (n^s abck)^3 + \\
& + \frac{1}{2} \sum_{i=2}^{n-1} i(i-1) b^{n(i-2)} h^{n-1-i} (n^s abck)^2 + \frac{1}{2} \sum_{i=0}^{n-3} (n-1-i)(n-2-i) b^{ni} h^{n-3-i} (n^s abck)^2 - \sum_{i=1}^{n-2} i(n-1-i) b^{n(i-1)} h^{n-2-i} (n^s abck)^2 +
\end{aligned}$$

$$\begin{aligned}
& + \sum_{i=1}^{n-1} i b^{n(i-1)} h^{n-1-i} n^s abck - \sum_{i=0}^{n-2} (n-1-i) b^{ni} h^{n-2-i} n^s abck + \sum_{i=0}^{n-1} h^{n-1-i} b^{ni} \\
& \equiv a^{n(n-1)} + na^{n(n-2)} n^s abck + \frac{n(n-1)}{2} a^{n(n-3)} (n^s abck)^2 + \frac{n(n-1)(n-2)}{6} a^{n(n-4)} (n^s abck)^3 - \frac{n}{4} a^{n(n-5)} (n^s abck)^4 \pmod{n^{4s+2}} \quad (\forall 5s \geq 4s+2, s \geq 2) \\
& \Rightarrow \frac{1}{24} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3) h^{n-1-i} b^{n(i-4)} (n^s abck)^4 + \frac{1}{24} \sum_{i=0}^{n-5} (n-1-i)(n-2-i)(n-3-i)(n-4-i) h^{n-5-i} b^{ni} (n^s abck)^4 + \\
& + \frac{1}{6} \sum_{i=1}^{n-4} i(1+i)(2+i)(3+i) h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \frac{n}{6} \sum_{i=1}^{n-4} [3i(i+1)(i+2) + 3i(i+1) + 2i] h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \\
& - \frac{n}{6} \sum_{i=3}^{n-2} i(i-1)(i-2) h^{n-2-i} b^{n(i-3)} (n^s abck)^4 + \frac{1}{6} \sum_{i=3}^{n-2} (1+i)i(i-1)(i-2) h^{n-2-i} b^{n(i-3)} (n^s abck)^4 - \frac{n}{4} \sum_{i=2}^{n-3} i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 + \\
& + \frac{1}{4} \sum_{i=2}^{n-3} (2+i)(1+i)i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 - \frac{n}{2} \sum_{i=2}^{n-3} (1+i)i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 + \\
& + \frac{1}{2} \sum_{i=1}^{n-3} i(n-1-i)(n-2-i) h^{n-3-i} b^{n(i-1)} (n^s abck)^3 - \frac{1}{2} \sum_{i=2}^{n-2} i(i-1)(n-1-i) h^{n-2-i} b^{n(i-2)} (n^s abck)^3 - \\
& - \frac{1}{6} \sum_{i=0}^{n-4} (n-1-i)(n-2-i)(n-3-i) h^{n-4-i} b^{ni} (n^s abck)^3 + \frac{1}{6} \sum_{i=3}^{n-1} i(i-1)(i-2) h^{n-1-i} b^{n(i-3)} (n^s abck)^3 + \\
& + \frac{1}{2} \sum_{i=2}^{n-1} i(i-1) b^{n(i-2)} h^{n-1-i} (n^s abck)^2 + \frac{1}{2} \sum_{i=0}^{n-3} (n-1-i)(n-2-i) b^{ni} h^{n-3-i} (n^s abck)^2 - \sum_{i=1}^{n-2} i(n-1-i) b^{n(i-1)} h^{n-2-i} (n^s abck)^2 + \\
& + \sum_{i=1}^{n-1} i b^{n(i-1)} h^{n-1-i} n^s abck - \sum_{i=0}^{n-2} (n-1-i) b^{ni} h^{n-2-i} n^s abck + \sum_{i=0}^{n-1} h^{n-1-i} b^{ni} \\
& \equiv a^{n(n-1)} + na^{n(n-2)} n^s abck + \frac{n(n-1)}{2} a^{n(n-3)} (n^s abck)^2 + \frac{n(n-1)(n-2)}{6} a^{n(n-4)} (n^s abck)^3 - \frac{n}{4} a^{n(n-5)} (n^s abck)^4 \pmod{n^{4s+2}} \\
& \Rightarrow \frac{1}{24} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3) h^{n-1-i} b^{n(i-4)} (n^s abck)^4 + \frac{1}{24} \sum_{i=0}^{n-5} (n-1-i)(n-2-i)(n-3-i)(n-4-i) h^{n-5-i} b^{ni} (n^s abck)^4 + \\
& + \frac{1}{6} \sum_{i=1}^{n-4} i(1+i)(2+i)(3+i) h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \frac{n}{2} \sum_{i=1}^{n-4} i(i+1)(i+2) h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \frac{n}{2} \sum_{i=1}^{n-4} i(i+1) h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \\
& - \frac{n}{3} \sum_{i=1}^{n-4} i h^{n-4-i} b^{n(i-1)} (n^s abck)^4 - \frac{n}{6} \sum_{i=3}^{n-2} i(i-1)(i-2) h^{n-2-i} b^{n(i-3)} (n^s abck)^4 + \frac{1}{6} \sum_{i=3}^{n-2} (1+i)i(i-1)(i-2) h^{n-2-i} b^{n(i-3)} (n^s abck)^4 - \\
& - \frac{n}{4} \sum_{i=2}^{n-3} i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 + \frac{1}{4} \sum_{i=2}^{n-3} (2+i)(1+i)i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 - \frac{n}{2} \sum_{i=2}^{n-3} (1+i)i(i-1) h^{n-3-i} b^{n(i-2)} (n^s abck)^4 + \\
& + \frac{1}{2} \sum_{i=1}^{n-3} i(n-1-i)(n-2-i) h^{n-3-i} b^{n(i-1)} (n^s abck)^3 - \frac{1}{2} \sum_{i=2}^{n-2} i(i-1)(n-1-i) h^{n-2-i} b^{n(i-2)} (n^s abck)^3 - \\
& - \frac{1}{6} \sum_{i=0}^{n-4} (n-1-i)(n-2-i)(n-3-i) h^{n-4-i} b^{ni} (n^s abck)^3 + \frac{1}{6} \sum_{i=3}^{n-1} i(i-1)(i-2) h^{n-1-i} b^{n(i-3)} (n^s abck)^3 + \\
& + \frac{1}{2} \sum_{i=2}^{n-1} i(i-1) b^{n(i-2)} h^{n-1-i} (n^s abck)^2 + \frac{1}{2} \sum_{i=0}^{n-3} (n-1-i)(n-2-i) b^{ni} h^{n-3-i} (n^s abck)^2 - \sum_{i=1}^{n-2} i(n-1-i) b^{n(i-1)} h^{n-2-i} (n^s abck)^2 + \\
& + \sum_{i=1}^{n-1} i b^{n(i-1)} h^{n-1-i} n^s abck - \sum_{i=0}^{n-2} (n-1-i) b^{ni} h^{n-2-i} n^s abck + \sum_{i=0}^{n-1} h^{n-1-i} b^{ni} \\
& \equiv a^{n(n-1)} + na^{n(n-2)} n^s abck + \frac{n(n-1)}{2} a^{n(n-3)} (n^s abck)^2 + \frac{n(n-1)(n-2)}{6} a^{n(n-4)} (n^s abck)^3 - \frac{n}{4} a^{n(n-5)} (n^s abck)^4 \pmod{n^{4s+2}}
\end{aligned}$$

$$\Rightarrow M(n^s abck)^4 + \frac{1}{2}B(n^s abck)^3 + \frac{1}{6}C(n^s abck)^3 + \frac{1}{2}D(n^s abck)^2 + E(n^s abck) + \frac{h^n - b^{n^2}}{h - b^n} \equiv F \pmod{n^{4s+2}}$$

$$\Rightarrow M(n^s abck)^4 + \frac{1}{2}B(n^s abck)^3 + \frac{1}{6}C(n^s abck)^3 + \frac{1}{2}D(n^s abck)^2 + E(n^s abck) + \frac{h^n - b^{n^2}}{h - b^n} - F \equiv 0 \pmod{n^{4s+2}} \quad (11)$$

$$\begin{aligned} \text{Với } M &= \frac{1}{24} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3)h^{n-1-i}b^{n(i-4)} + \frac{1}{24} \sum_{i=0}^{n-5} (n-1-i)(n-2-i)(n-3-i)(n-4-i)h^{n-5-i}b^{ni} + \\ &+ \frac{1}{6} \sum_{i=1}^{n-4} i(1+i)(2+i)(3+i)h^{n-4-i}b^{n(i-1)} - \frac{n}{2} \sum_{i=1}^{n-4} i(i+1)(i+2)h^{n-4-i}b^{n(i-1)} - \frac{n}{2} \sum_{i=1}^{n-4} i(i+1)h^{n-4-i}b^{n(i-1)} - \frac{n}{3} \sum_{i=1}^{n-4} ih^{n-4-i}b^{n(i-1)} - \\ &- \frac{n}{6} \sum_{i=3}^{n-2} i(i-1)(i-2)h^{n-2-i}b^{n(i-3)} + \frac{1}{6} \sum_{i=3}^{n-2} (1+i)i(i-1)(i-2)h^{n-2-i}b^{n(i-3)} - \frac{n}{4} \sum_{i=2}^{n-3} i(i-1)h^{n-3-i}b^{n(i-2)} + \frac{1}{4} \sum_{i=2}^{n-3} (2+i)(1+i)i(i-1)h^{n-3-i}b^{n(i-2)} - \\ &- \frac{n}{2} \sum_{i=2}^{n-3} (1+i)i(i-1)h^{n-3-i}b^{n(i-2)} \\ B &= \sum_{i=1}^{n-3} i(n-1-i)(n-2-i)h^{n-3-i}b^{n(i-1)} - \sum_{i=2}^{n-2} i(i-1)(n-1-i)h^{n-2-i}b^{n(i-2)} \\ C &= \sum_{i=3}^{n-1} i(i-1)(i-2)h^{n-1-i}b^{n(i-3)} - \sum_{i=0}^{n-4} (n-1-i)(n-2-i)(n-3-i)h^{n-4-i}b^{ni} \\ D &= \sum_{i=2}^{n-1} i(i-1)b^{n(i-2)}h^{n-1-i} + \sum_{i=0}^{n-3} (n-1-i)(n-2-i)b^{ni}h^{n-3-i} - 2 \sum_{i=1}^{n-2} i(n-1-i)b^{n(i-1)}h^{n-2-i} \\ E &= \sum_{i=1}^{n-1} ib^{n(i-1)}h^{n-1-i} - \sum_{i=0}^{n-2} (n-1-i)b^{ni}h^{n-2-i} \\ F &= a^{n(n-1)} + na^{n(n-2)}n^s abck + \frac{n(n-1)}{2}a^{n(n-3)}(n^s abck)^2 + \frac{n(n-1)(n-2)}{6}a^{n(n-4)}(n^s abck)^3 - \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \end{aligned}$$

Quy ước: Trong các biến đổi đồng dư thức từ đây trở đi, tác giả quy đồng mẫu số theo hướng sử dụng (7), (8), (9), (10) và không bỏ mẫu; điều này không làm mất bản chất phương trình đồng dư theo modulo n^f , $f \in \mathbb{N}^*$ (vì các mẫu là $m(h-b^n)^r \not\equiv 0 \pmod{n}$, $r=1, 2, 3, 4$; $m=1, 2, 3, 4, 6, 12$).

$$\begin{aligned} \text{Ta có: } * \quad M &= \frac{1}{24} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3)h^{n-1-i}b^{n(i-4)} + \frac{1}{24} \sum_{i=0}^{n-5} (n-1-i)(n-2-i)(n-3-i)(n-4-i)h^{n-5-i}b^{ni} + \\ &+ \frac{1}{6} \sum_{i=1}^{n-4} i(1+i)(2+i)(3+i)h^{n-4-i}b^{n(i-1)} - \frac{n}{2} \sum_{i=1}^{n-4} i(i+1)(i+2)h^{n-4-i}b^{n(i-1)} - \frac{n}{2} \sum_{i=1}^{n-4} i(i+1)h^{n-4-i}b^{n(i-1)} - \\ &- \frac{n}{6} \sum_{i=3}^{n-2} i(i-1)(i-2)h^{n-2-i}b^{n(i-3)} + \frac{1}{6} \sum_{i=3}^{n-2} (1+i)i(i-1)(i-2)h^{n-2-i}b^{n(i-3)} - \frac{n}{4} \sum_{i=2}^{n-3} i(i-1)h^{n-3-i}b^{n(i-2)} + \frac{1}{4} \sum_{i=2}^{n-3} (2+i)(1+i)i(i-1)h^{n-3-i}b^{n(i-2)} - \\ &- \frac{n}{2} \sum_{i=2}^{n-3} (1+i)i(i-1)h^{n-3-i}b^{n(i-2)} - \frac{n}{3} \sum_{i=1}^{n-4} ih^{n-4-i}b^{n(i-1)} \\ &= \frac{1}{24} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3)h^{n-1-i}b^{n(i-4)} + \frac{1}{24} \sum_{i=0}^{n-5} (n-1-i)(n-2-i)(n-3-i)(n-4-i)h^{n-5-i}b^{ni} + \\ &+ \frac{1}{6} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3)h^{n-1-i}b^{n(i-4)} - \frac{n}{2} \sum_{i=3}^{n-2} i(i-1)(i-2)h^{n-2-i}b^{n(i-3)} - \frac{n}{2} \sum_{i=2}^{n-3} i(i-1)h^{n-3-i}b^{n(i-2)} - \\ &- \frac{n}{6} \sum_{i=3}^{n-2} i(i-1)(i-2)h^{n-2-i}b^{n(i-3)} + \frac{1}{6} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3)h^{n-1-i}b^{n(i-4)} - \frac{n}{4} \sum_{i=2}^{n-3} i(i-1)h^{n-3-i}b^{n(i-2)} - \frac{n}{3} \sum_{i=1}^{n-4} ih^{n-4-i}b^{n(i-1)} \\ &+ \frac{1}{4} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3)h^{n-1-i}b^{n(i-4)} - \frac{n}{2} \sum_{i=3}^{n-2} i(i-1)(i-2)h^{n-2-i}b^{n(i-3)} \quad (\text{áp dụng mục B.1) trang 2}) \end{aligned}$$

$$\begin{aligned}
&= \frac{15}{24} \sum_{i=4}^{n-1} i(i-1)(i-2)(i-3)h^{n-1-i}b^{n(i-4)} + \frac{1}{24} \sum_{i=0}^{n-5} (n-1-i)(n-2-i)(n-3-i)(n-4-i)h^{n-5-i}b^{ni} - \frac{7n}{6} \sum_{i=3}^{n-2} i(i-1)(i-2)h^{n-2-i}b^{n(i-3)} \\
&\quad - \frac{3n}{4} \sum_{i=2}^{n-3} i(i-1)h^{n-3-i}b^{n(i-2)} - \frac{n}{3} \sum_{i=1}^{n-4} ih^{n-4-i}b^{n(i-1)} \\
&= \frac{15}{24} f^{(4)}(b^n) + \frac{1}{24} g^{(4)}(h) - \frac{7n}{6} r^{(3)}(b^n) - \frac{3n}{4} p''(b^n) - \frac{n}{3} h'(b^n) \text{ (áp dụng mục B.2) trang 2)} \\
\text{Với } f(x) &= \frac{x^n - h^n}{x - h}, g(x) = \frac{x^n - b^{n^2}}{x - b^n}, r(x) = \frac{x^{n-1} - h^{n-1}}{x - h}, p(x) = \frac{x^{n-2} - h^{n-2}}{x - h}, h(x) = \frac{x^{n-3} - h^{n-3}}{x - h} \\
f'(x) &= \frac{nx^{n-1}(x-h) - x^n + h^n}{(x-h)^2}, g'(x) = \frac{(n-1)x^n - nb^n x^{n-1} + b^{n^2}}{(x-b^n)^2}, r'(x) = \frac{(n-2)x^{n-1} - (n-1)hx^{n-2} + h^{n-1}}{(x-h)^2} \\
p'(x) &= \frac{(n-3)x^{n-2} - (n-2)hx^{n-3} + h^{n-2}}{(x-h)^2}, h'(x) = \frac{(n-4)x^{n-3} - (n-3)hx^{n-4} + h^{n-3}}{(x-h)^2} \\
f''(x) &= \frac{n(n-1)x^{n-2}(x-h)^2 - 2[nx^{n-1}(x-h) - x^n + h^n]}{(x-h)^3}, r''(x) = \frac{(n-1)(n-2)x^{n-3}(x-h)^2 - 2[(n-1)x^{n-2}(x-h) - x^{n-1} + h^{n-1}]}{(x-h)^3} \\
g''(x) &= \frac{n(n-1)x^{n-2}(x-b^n)^2 - 2[nx^{n-1}(x-b^n) - x^n + b^{n^2}]}{(x-b^n)^3}, p''(x) = \frac{(n-2)(n-3)x^{n-4}(x-h)^2 - 2[(n-2)x^{n-3}(x-h) - x^{n-2} + h^{n-2}]}{(x-h)^3} \\
r^{(3)}(x) &= \frac{(n-1)(n-2)(n-3)x^{n-4}(x-h)^3 - 3(n-1)(n-2)x^{n-3}(x-h)^2 + 6[(n-1)x^{n-2}(x-h) - x^{n-1} + h^{n-1}]}{(x-h)^4} \\
f^{(3)}(x) &= \frac{n(n-1)(n-2)x^{n-3}(x-h)^3 - 3n(n-1)x^{n-2}(x-h)^2 + 6[nx^{n-1}(x-h) - x^n + h^n]}{(x-h)^4} \\
f^{(4)}(x) &= \frac{n(n-1)(n-2)(n-3)x^{n-4}(x-h)^4 - 4n(n-1)(n-2)x^{n-3}(x-h)^3 + 12n(n-1)x^{n-2}(x-h)^2 - 24[nx^{n-1}(x-h) - x^n + h^n]}{(x-h)^5} \\
g^{(4)}(x) &= \frac{n(n-1)(n-2)(n-3)x^{n-4}(x-b^n)^4 - 4n(n-1)(n-2)x^{n-3}(x-b^n)^3 + 12n(n-1)x^{n-2}(x-b^n)^2 - 24[nx^{n-1}(x-b^n) - x^n + b^{n^2}]}{(x-b^n)^5}
\end{aligned}$$

$$\text{Ta có: } \frac{15}{24} f^{(4)}(b^n) \equiv \frac{5[3nb^{n(n-4)}(b^n - h)^2 - 4nb^{n(n-3)}(h - b^n) + 6nb^{n(n-2)}]}{4(h - b^n)^3} + \frac{15(h^n - b^{n^2})}{(h - b^n)^5} - \frac{15nb^{n(n-1)}}{(h - b^n)^4} \pmod{n^2}$$

$$\frac{1}{24} g^{(4)}(h) \equiv -\frac{3nh^{n-4}(b^n - h)^2 + 4nh^{n-3}(h - b^n) + 6nh^{n-2}}{12(h - b^n)^3} + \frac{h^n - b^{n^2}}{(h - b^n)^5} - \frac{nh^{n-1}}{(h - b^n)^4} \pmod{n^2}$$

$$-\frac{7n}{6} r^{(3)}(b^n) \equiv \frac{7nb^{n(n-4)}(b^n - h)^3 + 7nb^{n(n-3)}(b^n - h)^2 - 7n[-b^{n(n-2)}(b^n - h) - b^{n(n-1)} + h^{n-1}]}{(h - b^n)^4} \pmod{n^2}$$

$$-\frac{3n}{4} p''(b^n) \equiv \frac{9nb^{n(n-4)}(h - b^n)^2 + 3n[-2b^{n(n-3)}(h - b^n) + b^{n(n-2)} - h^{n-2}]}{2(h - b^n)^3} \pmod{n^2}$$

$$-\frac{n}{3} h'(b^n) \equiv \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h - b^n)^2} \pmod{n^2}$$

$$\Rightarrow \frac{15}{24} f^{(4)}(b^n) + \frac{1}{24} g^{(4)}(h) - \frac{7n}{6} r^{(3)}(b^n) - \frac{3n}{4} p''(b^n) - \frac{n}{3} h'(b^n)$$

$$\equiv \frac{5[3nb^{n(n-4)}(b^n - h)^2 - 4nb^{n(n-3)}(h - b^n) + 6nb^{n(n-2)}]}{4(h - b^n)^3} + \frac{15(h^n - b^{n^2})}{(h - b^n)^5} - \frac{15nb^{n(n-1)}}{(h - b^n)^4}$$

$$- \frac{3nh^{n-4}(b^n - h)^2 + 4nh^{n-3}(h - b^n) + 6nh^{n-2}}{12(h - b^n)^3} + \frac{h^n - b^{n^2}}{(h - b^n)^5} - \frac{nh^{n-1}}{(h - b^n)^4} + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h - b^n)^2} +$$

$$+ \frac{7nb^{n(n-4)}(b^n - h)^3 + 7nb^{n(n-3)}(b^n - h)^2 - 7n[b^{n(n-2)}(h - b^n) + h^{n-1} - b^{n(n-1)}]}{(h - b^n)^4}$$

$$+ \frac{9nb^{n(n-4)}(h - b^n)^2 + 3n[-2b^{n(n-3)}(h - b^n) + b^{n(n-2)} - h^{n-2}]}{2(h - b^n)^3}$$

$$\text{Vậy } M \equiv \frac{16(h^n - b^{n^2})}{(h - b^n)^5} + M_1 \pmod{n^2}.$$

Với

$$M_1 = \frac{7nb^{n(n-4)}(b^n - h)^3 + 7nb^{n(n-3)}(b^n - h)^2 - 7n[b^{n(n-2)}(h - b^n) + h^{n-1} - b^{n(n-1)}]}{(h - b^n)^4} + \frac{5[3nb^{n(n-4)}(b^n - h)^2 - 4nb^{n(n-3)}(h - b^n) + 6nb^{n(n-2)}]}{4(h - b^n)^3}$$

$$- \frac{3nh^{n-4}(b^n - h)^2 + 4nh^{n-3}(h - b^n) + 6nh^{n-2}}{12(h - b^n)^3} - \frac{nh^{n-1}}{(b^n - h)^4} + \frac{9nb^{n(n-4)}(b^n - h)^2 + 3n[-2b^{n(n-3)}(h - b^n) - b^{n(n-2)} + h^{n-2}]}{2(h - b^n)^3} +$$

$$+ \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(b^n - h)^2} - \frac{15nb^{n(n-1)}}{(b^n - h)^4}$$

$$* B = \sum_{i=1}^{n-3} i(n-1-i)(n-2-i)h^{n-3-i}b^{n(i-1)} - \sum_{i=2}^{n-2} i(i-1)(n-1-i)h^{n-2-i}b^{n(i-2)}$$

$$= \left[\sum_{i=1}^{n-3} i(i+1)(i+2)h^{n-3-i}b^{n(i-1)} - n \sum_{i=1}^{n-3} i(2i+3)h^{n-3-i}b^{n(i-1)} + n^2 \sum_{i=1}^{n-3} ih^{n-3-i}b^{n(i-1)} \right] + \left[\sum_{i=2}^{n-2} (i-1)i(i+1)h^{n-2-i}b^{n(i-2)} - n \sum_{i=2}^{n-2} i(i-1)h^{n-2-i}b^{n(i-2)} \right]$$

$$= \left[\sum_{i=1}^{n-3} i(i+1)(i+2)h^{n-3-i}b^{n(i-1)} - n \sum_{i=1}^{n-3} i(2i+3)h^{n-3-i}b^{n(i-1)} + n^2 \sum_{i=1}^{n-3} ih^{n-3-i}b^{n(i-1)} \right] + \left[\sum_{i=3}^{n-1} (i-2)(i-1)ih^{n-1-i}b^{n(i-3)} - n \sum_{i=2}^{n-2} i(i-1)h^{n-2-i}b^{n(i-2)} \right]$$

$$= \left\{ \left[\sum_{i=3}^{n-1} (i-2)(i-1)ih^{n-1-i}b^{n(i-3)} + \sum_{i=3}^{n-1} (i-2)(i-1)ih^{n-1-i}b^{n(i-3)} \right] - n \left[\sum_{i=2}^{n-2} (i-1)(2i+1)h^{n-2-i}b^{n(i-2)} + \sum_{i=2}^{n-2} i(i-1)h^{n-2-i}b^{n(i-2)} \right] \right\} +$$

$$+ n^2 \sum_{i=1}^{n-3} ih^{n-3-i}b^{n(i-1)} \text{ (áp dụng mục B.1), trang 2)}$$

$$= \left[2 \sum_{i=3}^{n-1} (i-2)(i-1)ih^{n-1-i}b^{n(i-3)} - n \sum_{i=2}^{n-2} (i-1)(3i+1)h^{n-2-i}b^{n(i-2)} \right] + n^2 \sum_{i=1}^{n-3} ih^{n-3-i}b^{n(i-1)}$$

$$= B^* + n^2 \sum_{i=1}^{n-3} ih^{n-3-i}b^{n(i-1)}$$

$$= B^* + n^2 p'(b^n) \text{ (áp dụng mục B.2), trang 2)}$$

$$= B^* + \frac{n^2[(n-3)b^{n(n-2)} - (n-2)hb^{n(n-3)} + h^{n-2}]}{(b^n - h)^2}$$

$$= B^* + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}]}{(b^n - h)^2} - \frac{n^3b^{n(n-3)}}{h - b^n}$$

$$\text{Với } B^* = 2 \sum_{i=3}^{n-1} (i-2)(i-1)ih^{n-1-i}b^{n(i-3)} - n \sum_{i=2}^{n-2} (i-1)(3i+1)h^{n-2-i}b^{n(i-2)}$$

$$= 2 \sum_{i=3}^{n-1} i(i-1)(i-2)h^{n-1-i}b^{n(i-3)} - n \left[3 \sum_{i=2}^{n-2} i(i-1)h^{n-2-i}b^{n(i-2)} + \sum_{i=2}^{n-2} (i-1)h^{n-2-i}b^{n(i-2)} \right]$$

$$= 2 \sum_{i=3}^{n-1} i(i-1)(i-2)h^{n-1-i}b^{n(i-3)} - n \left[3 \sum_{i=2}^{n-2} i(i-1)h^{n-2-i}b^{n(i-2)} + \sum_{i=1}^{n-3} ih^{n-3-i}b^{n(i-1)} \right] \text{ (áp dụng mục B.1), trang 2)}$$

$$= 2f^{(3)}(b^n) - n[3r''(b^n) + p'(b^n)] \text{ (áp dụng mục B.2), trang 2)}$$

$$\begin{aligned}
\text{suy ra } B^* &= 2 \frac{n(n-1)(n-2)b^{n(n-3)}(b^n-h)^3 - 3n(n-1)b^{n(n-2)}(b^n-h)^2 + 6[nb^{n(n-1)}(b^n-h) - b^{n^2} + h^n]}{(b^n-h)^4} - \\
&- \frac{3n\{(n-1)(n-2)b^{n(n-3)}(b^n-h)^2 - 2[(n-2)b^{n(n-1)} - (n-1)hb^{n(n-2)} + h^{n-1}]\}}{(b^n-h)^3} - \frac{n[(n-3)b^{n(n-2)} - (n-2)hb^{n(n-3)} + h^{n-2}]}{(b^n-h)^2} \\
&= 2 \frac{[n(n-1)^2b^{n(n-2)} - n(n-1)(n-2)hb^{n(n-3)}](b^n-h)^2 - 4[n(n-1)b^{n(n-1)} - n(n-1)hb^{n(n-2)}](b^n-h) + 6[(n-1)b^{n^2} - nb^{n(n-1)} + h^n]}{(b^n-h)^4} - \\
&- \frac{3n\{[(n-1)(n-2)b^{n(n-2)} - (n-1)(n-2)hb^{n(n-3)}](b^n-h) - 2[(n-2)b^{n(n-1)} - (n-1)hb^{n(n-2)} + h^{n-1}]\}}{(b^n-h)^3} - \\
&- \frac{n[(n-3)b^{n(n-2)} - (n-2)hb^{n(n-3)} + h^{n-2}]}{(b^n-h)^2} \\
&= \frac{2[(n^3-2n^2+n)b^{n(n-2)} - (n^3-3n^2+2n)hb^{n(n-3)}](b^n-h)^2 - 8n^2b^{n(n-2)}(b^n-h)^2 + 8nb^{n(n-2)}(b^n-h)^2 + 12[nb^{n(n-1)}(b^n-h) - b^{n^2} + h^n]}{(b^n-h)^4} - \\
&+ \frac{3\{[(n^3-3n^2+2n)b^{n(n-2)} - (n^3-3n^2+2n)hb^{n(n-3)}](b^n-h) - 2[n^2b^{n(n-2)}(b^n-h) + nb^{n(n-2)} - 2nb^{n(n-1)} + nh^{n-1}]\}}{(h-b^n)^3} - \\
&- \frac{[(n^2-3n)b^{n(n-2)} - (n^2-2n)hb^{n(n-3)} + nh^{n-2}]}{(b^n-h)^2} \\
\Rightarrow B^* &= \frac{12(h^n-b^{n^2})}{(b^n-h)^4} + \frac{2[nb^{n(n-2)} - 2nb^{n(n-3)}](b^n-h)^2 + 8nb^{n(n-2)}(h-b^n)^2 - 12nb^{n(n-1)}(h-b^n)}{(b^n-h)^4} - \\
&+ \frac{3\{[2nb^{n(n-2)} - 2nb^{n(n-3)}](b^n-h) - 2[-2nb^{n(n-1)} + nb^{n(n-2)} + nh^{n-1}]\}}{(h-b^n)^3} - \frac{[-3nb^{n(n-2)} + 2nb^{n(n-3)} + nh^{n-2}]}{(b^n-h)^2} + \\
&+ \frac{2n^2b^{n(n-2)}}{(h-b^n)^2} - \frac{2n^2hb^{n(n-3)}}{(h-b^n)^2} + \left[\frac{n^3hb^{n(n-3)}}{(h-b^n)^2} - \frac{n^3b^{n(n-2)}}{(h-b^n)^2} \right] \\
\Rightarrow B^* &= \frac{12(h^n-b^{n^2})}{(b^n-h)^4} + \frac{[10nb^{n(n-2)} - 4nb^{n(n-3)}](h-b^n) - 12nb^{n(n-1)}}{(h-b^n)^3} - \frac{[-3nb^{n(n-2)} + 2nb^{n(n-3)} + nh^{n-2}](h-b^n)}{(h-b^n)^3} \\
&+ \frac{3\{[-2nb^{n(n-2)} + 2nb^{n(n-3)}](h-b^n) - 2[-2nb^{n(n-1)} + nb^{n(n-2)} + nh^{n-1}]\}}{(h-b^n)^3} + \frac{2n^2b^{n(n-2)}}{(h-b^n)^2} - \frac{2n^2hb^{n(n-3)}}{(h-b^n)^2} + \frac{n^3b^{n(n-3)}}{h-b^n} \\
\Rightarrow B^* &= \frac{12(h^n-b^{n^2})}{(b^n-h)^4} + \frac{[10nb^{n(n-2)} - 4nb^{n(n-3)}]a^n - 12nb^{n(n-1)}}{(h-b^n)^3} - \frac{n[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}]a^n}{(h-b^n)^3} + \\
&+ \frac{[-6nb^{n(n-2)} + 6nb^{n(n-3)}]a^n + 12nb^{n(n-1)} - 6nb^{n(n-2)} - 6nh^{n-1}}{(h-b^n)^3} + \frac{2n^2b^{n(n-2)}}{(h-b^n)^2} - \frac{2n^2hb^{n(n-3)}}{(h-b^n)^2} - \frac{[-3nb^{n(n-2)} + 2nb^{n(n-3)} + nh^{n-2}]2n^s abck}{(h-b^n)^3} + \\
&+ \frac{[10nb^{n(n-2)} - 4nb^{n(n-3)}]2n^s abck}{(h-b^n)^3} + \frac{3[-2nb^{n(n-2)} + 2nb^{n(n-3)}]2n^s abck}{(h-b^n)^3} + \frac{n^3b^{n(n-3)}}{h-b^n} \text{ (vì } h-b^n = a^n + 2n^s abck \text{)} \\
\Rightarrow B^* &= \frac{7nb^{n(n-2)}a^n - 6nh^{n-1} - nh^{n-2}a^n - 6nb^{n(n-2)}}{(h-b^n)^3} + \frac{12(h^n-b^{n^2})}{(h-b^n)^4} - \frac{2n^2b^{n(n-3)}}{h-b^n} + \frac{n^3b^{n(n-3)}}{h-b^n} + \\
&+ \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}]}{(b^n-h)^2} + \frac{14nb^{n(n-2)}n^s abck - 2nh^{n-2}n^s abck}{(h-b^n)^3} \\
\text{Vậy } B &= B_1 + B_2 + B_3 = \frac{7nb^{n(n-2)}a^n - 6nh^{n-1} - nh^{n-2}a^n - 6nb^{n(n-2)}}{(h-b^n)^3} + \frac{12(h^n-b^{n^2})}{(h-b^n)^4} - \frac{2n^2b^{n(n-3)}}{h-b^n}
\end{aligned}$$

$$+ \frac{14nb^{n(n-2)} n^s abck - 2nh^{n-2} n^s abck}{(h-b^n)^3} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}]}{(b^n-h)^2}$$

Với $B_1 = \frac{7nb^{n(n-2)} a^n - 6nh^{n-1} - nh^{n-2} a^n - 6nhb^{n(n-2)}}{(h-b^n)^3} + \frac{12(h^n - b^{n^2})}{(h-b^n)^4}$ và

$$B_2 = -\frac{2n^2 b^{n(n-3)}}{h-b^n} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}]}{(b^n-h)^2}, B_3 = \frac{14nb^{n(n-2)} n^s abck - 2nh^{n-2} n^s abck}{(h-b^n)^3}$$

$$*C = \sum_{i=3}^{n-1} i(i-1)(i-2)h^{n-1-i}b^{n(i-3)} - \sum_{i=0}^{n-4} (n-1-i)(n-2-i)(n-3-i)h^{n-4-i}b^{ni}$$

= $f^{(3)}(b^n) - g^{(3)}(h)$ (áp dụng mục B.2), trang 2)

$$= \frac{n(n-1)(n-2)b^{n(n-3)}(b^n-h)^3 - 3n(n-1)b^{n(n-2)}(b^n-h)^2 + 6[nb^{n(n-1)}(b^n-h) - b^{n^2} + h^n]}{(b^n-h)^4}$$

$$- \frac{n(n-1)(n-2)h^{n-3}(h-b^n)^3 - 3n(n-1)h^{n-2}(h-b^n)^2 + 6[nh^{n-1}(h-b^n) - h^n + b^{n^2}]}{(h-b^n)^4}$$

$$= \frac{[(n^3 - 6n^2 + 5n)b^{n(n-2)} - (n^3 - 3n^2 + 2n)hb^{n(n-3)}](b^n-h)^2 + 6[nb^{n(n-1)}(b^n-h) - b^{n^2} + h^n]}{(b^n-h)^4}$$

$$- \frac{[(n^3 - 6n^2 + 5n)h^{n-2} - (n^3 - 3n^2 + 2n)b^n h^{n-3}](h-b^n)^2 + 6[nh^{n-1}(b^n-h) - h^n + b^{n^2}]}{(h-b^n)^4}$$

$$= \frac{12(h^n - b^{n^2})}{(h-b^n)^4} + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}](h-b^n)^2 - 6nb^{n(n-1)}(h-b^n) + [-5nh^{n-2} + 2nb^n h^{n-3}](h-b^n)^2 - 6nh^{n-1}(h-b^n)}{(h-b^n)^4} +$$

$$+ \frac{n^3 b^n h[h^{n-4} - b^{n(n-4)}] - n^3[h^{n-2} - b^{n(n-2)}]}{(h-b^n)^2} + \frac{6n^2[h^{n-2} - b^{n(n-2)}] - 3n^2 b^n h[h^{n-4} - b^{n(n-4)}]}{(h-b^n)^2}$$

$$= \frac{[-5nh^{n-2} + 2nb^n h^{n-3}](h-b^n) - 6nh^{n-1} + [5nb^{n(n-2)} - 2nhb^{n(n-3)}](h-b^n) - 6nb^{n(n-1)}}{(h-b^n)^3} + \frac{12(h^n - b^{n^2})}{(h-b^n)^4} +$$

$$+ \frac{n^3 b^n h[h^{n-4} - b^{n(n-4)}] - n^3[h^{n-2} - b^{n(n-2)}]}{(h-b^n)^2} + \frac{6n^2[h^{n-2} - b^{n(n-2)}] - 3n^2 b^n h[h^{n-4} - b^{n(n-4)}]}{(h-b^n)^2}$$

$$= \frac{[-5nh^{n-2} + 2nb^n h^{n-3}](2n^s abck) + [5nb^{n(n-2)} - 2nhb^{n(n-3)}](2n^s abck)}{(h-b^n)^3} + \frac{n^3 b^n h[h^{n-4} - b^{n(n-4)}] - n^3[h^{n-2} - b^{n(n-2)}]}{(h-b^n)^2} +$$

$$+ \frac{[-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1} + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)}}{(h-b^n)^3} + \frac{12(h^n - b^{n^2})}{(h-b^n)^4} +$$

$$+ \frac{6n^2[h^{n-2} - b^{n(n-2)}] - 3n^2 b^n h[h^{n-4} - b^{n(n-4)}]}{(h-b^n)^2} \text{ (vì } h-b^n = a^n + 2n^s abck \text{)}$$

$$\text{Vậy } C = C_1 + C_2 + C_3 = \frac{12(h^n - b^{n^2})}{(h-b^n)^4} - \frac{5na^n[h^{n-2} - b^{n(n-2)}] + 2na^n b^n h[h^{n-4} - b^{n(n-4)}] + 6n[h^{n-1} + b^{n(n-1)}]}{(h-b^n)^3} +$$

$$+ \frac{6n^2([h^{n-2} - b^{n(n-2)}]) - 3n^2 b^n h[h^{n-4} - b^{n(n-4)}]}{(h-b^n)^2} + \frac{5n[h^{n-2} - b^{n(n-2)}] + 2nb^n h[h^{n-4} - b^{n(n-4)}]}{(h-b^n)^3} (2n^s abck) +$$

$$+ \frac{n^3 b^n h[h^{n-4} - b^{n(n-4)}] - n^3[h^{n-2} - b^{n(n-2)}]}{(h-b^n)^2}, \text{ với } C_2 = \frac{6n^2([h^{n-2} - b^{n(n-2)}]) - 3n^2 b^n h[h^{n-4} - b^{n(n-4)}]}{(h-b^n)^2}$$

$$C_1 = \frac{12(h^n - b^{n^2})}{(h-b^n)^4} - \frac{5na^n[h^{n-2} - b^{n(n-2)}] + 2na^n b^n h[h^{n-4} - b^{n(n-4)}] + 6n[h^{n-1} + b^{n(n-1)}]}{(h-b^n)^3}$$

$$\begin{aligned}
C_3 &= \frac{5n[h^{n-2} - b^{n(n-2)}] + 2nb^n h[h^{n-4} - b^{n(n-4)}]}{(h-b^n)^3} (2n^s abck) + \frac{n^3 b^n h[h^{n-4} - b^{n(n-4)}] - n^3[h^{n-2} - b^{n(n-2)}]}{(h-b^n)^2} \\
&* D = \sum_{i=2}^{n-1} i(i-1)b^{n(i-2)}h^{n-1-i} + \sum_{i=0}^{n-3} (n-1-i)(n-2-i)b^{ni}h^{n-3-i} - 2\sum_{i=1}^{n-2} i(n-1-i)b^{n(i-1)}h^{n-2-i} \\
&= \sum_{i=2}^{n-1} i(i-1)b^{n(i-2)}h^{n-1-i} + \left[(n^2-3n)\sum_{i=0}^{n-3} b^{ni}h^{n-3-i} - 2n\sum_{i=1}^{n-3} ib^{ni}h^{n-3-i} + \sum_{i=0}^{n-3} (i+1)(i+2)h^{n-3-i}b^{ni} \right] - 2n\sum_{i=1}^{n-2} ib^{n(i-1)}h^{n-2-i} + \\
&+ 2\sum_{i=1}^{n-2} i(1+i)b^{n(i-1)}h^{n-2-i} = \sum_{i=2}^{n-1} i(i-1)b^{n(i-2)}h^{n-1-i} + (n^2-3n)\sum_{i=0}^{n-3} b^{ni}h^{n-3-i} - 2nb^n\sum_{i=1}^{n-3} ib^{n(i-1)}h^{n-3-i} + \\
&+ \sum_{i=2}^{n-1} i(i-1)b^{n(i-2)}h^{n-1-i} - 2n\sum_{i=1}^{n-2} ib^{n(i-1)}h^{n-2-i} + 2\sum_{i=2}^{n-1} i(i-1)b^{n(i-2)}h^{n-1-i} \text{ (áp dụng mục B.1), trang 2)} \\
&= 4\sum_{i=2}^{n-1} i(i-1)b^{n(i-2)}h^{n-1-i} + (n^2-3n)\sum_{i=0}^{n-3} b^{ni}h^{n-3-i} - 2nb^n\sum_{i=1}^{n-3} ib^{n(i-1)}h^{n-3-i} - 2n\sum_{i=1}^{n-2} ib^{n(i-1)}h^{n-2-i} \\
&= 4f''(b^n) + \frac{(n^2-3n)[h^{n-2} - b^{n(n-2)}]}{h-b^n} - 2nb^n p'(b^n) - 2nr'(b^n) \text{ (áp dụng mục B.2), trang 2)} \\
&= 4 \frac{[n(n-1)b^{n(n-1)} - n(n-1)hb^{n(n-2)}](b^n-h) - 2[(n-1)b^{n^2} - nhb^{n(n-1)} + h^n]}{(b^n-h)^3} + \frac{(n^2-3n)[h^{n-2} - b^{n(n-2)}]}{h-b^n} - \\
&- \frac{2n[(n-3)b^{n(n-1)} - (n-2)hb^{n(n-2)} + h^{n-2}b^n]}{(b^n-h)^2} - \frac{2n[(n-2)b^{n(n-1)} - (n-1)hb^{n(n-2)} + h^{n-1}]}{(b^n-h)^2} \\
&= \frac{[4n^2b^{n(n-1)} - 4nb^{n(n-1)} - 4n^2hb^{n(n-2)} + 4nhb^{n(n-2)}](b^n-h) - 8[nb^{n(n-1)}(b^n-h) - b^{n^2} + h^n]}{(b^n-h)^3} + \frac{(n^2-3n)[h^{n-2} - b^{n(n-2)}]}{h-b^n} - \\
&- \frac{2n^2b^{n(n-1)} - 6nb^{n(n-1)} - 2n^2hb^{n(n-2)} + 4nhb^{n(n-2)} + 2nh^{n-2}b^n}{(b^n-h)^2} - \frac{[2n^2b^{n(n-1)} - 4nb^{n(n-1)} - 2n^2hb^{n(n-2)} + 2nhb^{n(n-2)} + 2nh^{n-1}]}{(b^n-h)^2} \\
\text{Vậy } D &= \frac{8(h^n - b^{n^2})}{(h-b^n)^3} - \frac{2nb^{n(n-2)}h + 2nb^{n(n-1)} + 2nh^{n-2}b^n + 2nh^{n-1}}{(h-b^n)^2} + \frac{(n^2-3n)[h^{n-2} - b^{n(n-2)}]}{h-b^n} \\
&* E = \sum_{i=1}^{n-1} ib^{n(i-1)}h^{n-1-i} - \sum_{i=0}^{n-2} (n-1-i)b^{ni}h^{n-2-i} = f'(b^n) - g'(h) \\
&= f'(b^n) - g'(h) \text{ (áp dụng mục B.2), trang 2)} \\
&= \frac{(n-1)b^{n^2} - nhb^{n(n-1)} + h^n}{(h-b^n)^2} - \frac{(n-1)h^n - nb^n h^{n-1} + b^{n^2}}{(h-b^n)^2} \\
&= -\frac{nh^{n-1}(h-b^n) + nb^{n(n-1)}(h-b^n)}{(h-b^n)^2} + \frac{2(h^n - b^{n^2})}{(b^n-h)^2} = -\frac{n[h^{n-1} + b^{n(n-1)}]}{h-b^n} + \frac{2(h^n - b^{n^2})}{(b^n-h)^2} \\
&* F = a^{n(n-1)} + na^{n(n-2)}n^s abck + \frac{n(n-1)}{2}a^{n(n-3)}(n^s abck)^2 + \frac{n(n-1)(n-2)}{6}a^{n(n-4)}(n^s abck)^3 - \frac{n}{4}a^{n(n-5)}(n^s abck)^4 = \\
&= \frac{a^{n(n-1)} + na^{n(n-2)}n^s abck}{h-b^n} (a^n + 2n^s abck) + \frac{n(n-1)}{2}a^{n(n-3)}(n^s abck)^2 + \frac{n(n-1)(n-2)}{6}a^{n(n-4)}(n^s abck)^3 - \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \\
&= \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{[2a^{n(n-1)}n^s abck + 2na^{n(n-2)}(n^s abck)^2](a^n + 2n^s abck)}{(h-b^n)^2} + \frac{n(n-1)a^{n(n-3)}(n^s abck)^2(a^n + 2n^s abck)^2}{2(h-b^n)^2} + \\
&+ \frac{n(n-1)(n-2)}{6}a^{n(n-4)}(n^s abck)^3 \text{ (vì } h-b^n = a^n + 2n^s abck)
\end{aligned}$$

$$\begin{aligned}
&= \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck + 2na^{n(n-1)}(n^s abck)^2 + 4a^{n(n-1)}(n^s abck)^2 + 4na^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n}{4}a^{n(n-5)}(n^s abck)^4 + \\
&+ \frac{\frac{n(n-1)}{2}a^{n(n-1)}(n^s abck)^2 + 2n(n-1)a^{n(n-2)}(n^s abck)^3 + 2n(n-1)a^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \frac{n(n-1)(n-2)}{6}a^{n(n-4)}(n^s abck)^3 \\
&= \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck}{(h-b^n)^2} + \frac{4a^{n(n-1)}(n^s abck)^2}{(h-b^n)^2} + \frac{2na^{n(n-1)}(n^s abck)^2 + \frac{n(n-1)}{2}a^{n(n-1)}(n^s abck)^2}{(h-b^n)^2} + \\
&+ \frac{2na^{n(n-2)}(n^s abck)^3 + 2n^2a^{n(n-2)}(n^s abck)^3 + 2n(n-1)a^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \frac{n(n-1)(n-2)}{6}a^{n(n-4)}(n^s abck)^3 \\
&= \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck}{(h-b^n)^2} + \frac{4a^{n(n-1)}(n^s abck)^2(a^n + 2n^s abck)}{(h-b^n)^3} + \frac{3na^{n(n-1)}(n^s abck)^2 + n^2a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \\
&+ \frac{2n^2a^{n(n-2)}(n^s abck)^3 + 2n(n-1)a^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \frac{12na^{n(n-2)}(n^s abck)^3}{6(h-b^n)^2} + \frac{n(n-1)(n-2)}{6}a^{n(n-4)}(n^s abck)^3 - \\
&- \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \\
&= \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck}{(h-b^n)^2} + \frac{4a^{n^2}(n^s abck)^2}{(h-b^n)^3} + \frac{8a^{n(n-1)}(n^s abck)^3(a^n + 2n^s abck)}{(h-b^n)^4} + \\
&+ \frac{3na^{n(n-1)}(n^s abck)^2 + n^2a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \frac{2n^2a^{n(n-2)}(n^s abck)^3 + 2n(n-1)a^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} - \frac{n}{4}a^{n(n-5)}(n^s abck)^4 + \\
&+ \frac{12na^{n(n-2)}(a^n + 2n^s abck)(n^s abck)^3}{6(h-b^n)^3} + \left[\frac{n^2(n-3)}{6}a^{n(n-4)}(n^s abck)^3 + \frac{2n}{6}a^{n(n-4)}(n^s abck)^3 \right] \left(\text{vì } h-b^n = a^n + 2n^s abck \right) \\
&= \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck}{(h-b^n)^2} + \frac{4a^{n^2}(n^s abck)^2}{(h-b^n)^3} + \frac{8a^{n(n-1)}(n^s abck)^3(a^n + 2n^s abck)}{(h-b^n)^4} + \frac{2n}{6}a^{n(n-4)}(n^s abck)^3 + \\
&+ \frac{3na^{n(n-1)}(n^s abck)^2 + n^2a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \frac{2n^2a^{n(n-2)}(n^s abck)^3 + 2n(n-1)a^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \frac{n^2(n-3)}{6}a^{n(n-4)}(n^s abck)^3 + \\
&+ \frac{12na^{n(n-2)}(a^n + 2n^s abck)(n^s abck)^3}{6(h-b^n)^3} - \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \\
&= \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck}{(h-b^n)^2} + \frac{4a^{n^2}(n^s abck)^2}{(h-b^n)^3} + \frac{8a^{n^2}(n^s abck)^3}{(h-b^n)^4} + \frac{3na^{n(n-1)}(n^s abck)^2 + n^2a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \\
&+ \frac{12na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \frac{24na^{n(n-2)}(n^s abck)^4}{6(h-b^n)^3} + \left\{ \frac{2na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \frac{2na^{n(n-4)}(n^s abck)^3[(a^n + 2n^s abck)^3 - a^{3n}]}{6(h-b^n)^3} \right\} + \\
&+ \frac{16a^{n(n-1)}(n^s abck)^4}{(h-b^n)^4} + \frac{n^2(n-3)}{6}a^{n(n-4)}(n^s abck)^3 + \frac{2n^2a^{n(n-2)}(n^s abck)^3 + 2n(n-1)a^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} - \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \\
&= \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck}{(h-b^n)^2} + \frac{4a^{n^2}(n^s abck)^2}{(h-b^n)^3} + \frac{8a^{n^2}(n^s abck)^3}{(h-b^n)^4} + \frac{3na^{n(n-1)}(n^s abck)^2 + n^2a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \\
&+ \frac{12na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \frac{24na^{n(n-2)}(n^s abck)^4}{6(h-b^n)^3} + \frac{2na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \frac{2na^{n(n-4)}(n^s abck)^3[(a^n + 2n^s abck)^3 - a^{3n}]}{6(h-b^n)^3} +
\end{aligned}$$

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$$+ \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \frac{2n^2 a^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} + \frac{16a^{n(n-1)}(n^s abck)^4}{(h-b^n)^4} - \frac{n^2}{2} a^{n(n-4)}(n^s abck)^3 +$$

$$+ \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} + \frac{n^3}{6} a^{n(n-4)}(n^s abck)^3 - \frac{n}{4} a^{n(n-5)}(n^s abck)^4 \pmod{n^{4s+2}}$$

$$\text{Vậy } F \equiv F_1 + F_2 + F_3 \pmod{n^{4s+2}}$$

$$\text{với } F_1 = \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck}{(h-b^n)^2} + \frac{4a^{n^2}(n^s abck)^2}{(h-b^n)^3} + \frac{8a^{n^2}(n^s abck)^3}{(h-b^n)^4} + \frac{3na^{n(n-1)}(n^s abck)^2 + n^2 a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} +$$

$$+ \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3}, F_2 = \frac{2n^2 a^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} + \frac{16a^{n(n-1)}(n^s abck)^4}{(h-b^n)^4} - \frac{n^2}{2} a^{n(n-4)}(n^s abck)^3$$

$$F_3 = \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} + \frac{n^3}{6} a^{n(n-4)}(n^s abck)^3 - \frac{n}{4} a^{n(n-5)}(n^s abck)^4$$

$$\text{Ta có } T + Q = \left[\frac{1}{2} B_1(n^s abck)^3 + \frac{1}{6} C_1(n^s abck)^3 + \frac{1}{2} D(n^s abck)^2 + E(n^s abck) + \frac{h^n - b^{n^2}}{h - b^n} - F_1 \right] - F_3$$

$$+ \left[M(n^s abck)^4 + \frac{1}{2} B_2(n^s abck)^3 + \frac{1}{6} C_2(n^s abck)^3 - F_2 + \frac{1}{6} C_3(n^s abck)^3 + \frac{1}{2} B_3(n^s abck)^3 \right] \quad \text{với}$$

$$Q = M(n^s abck)^4 + \frac{1}{2} B_2(n^s abck)^3 + \frac{1}{2} B_3(n^s abck)^3 + \frac{1}{6} C_2(n^s abck)^3 + \frac{1}{6} C_3(n^s abck)^3 - F_2 \text{ và}$$

$$T = P - F_3, P = \frac{1}{2} B_1(n^s abck)^3 + \frac{1}{6} C_1(n^s abck)^3 + \frac{1}{2} D(n^s abck)^2 + E(n^s abck) + \frac{h^n - b^{n^2}}{h - b^n} - F_1$$

$$\text{Từ (11) suy ra } T + Q \equiv 0 \pmod{n^{4s+2}} \quad (12)$$

Dựa vào các kết quả biến đổi đồng nhất và đồng dư thức của các biểu thức:

$$B_1 = \frac{7nb^{n(n-2)}a^n - 6nh^{n-1} - nh^{n-2}a^n - 6nhb^{n(n-2)}}{(h-b^n)^3} + \frac{12(h^n - b^{n^2})}{(h-b^n)^4},$$

$$C_1 = \frac{12(h^n - b^{n^2})}{(h-b^n)^4} - \frac{5na^n[h^{n-2} - b^{n(n-2)}] + 2na^n b^n h[h^{n-4} - b^{n(n-4)}] + 6n[h^{n-1} + b^{n(n-1)}]}{(h-b^n)^3},$$

$$D = \frac{8(h^n - b^{n^2})}{(h-b^n)^3} - \frac{2nb^{n(n-2)}h + 2nb^{n(n-1)} + 2nh^{n-2}b^n + 2nh^{n-1}}{(h-b^n)^2} + \frac{(n^2 - 3n)[h^{n-2} - b^{n(n-2)}]}{h-b^n}, E = -\frac{n[h^{n-1} + b^{n(n-1)}]}{h-b^n} + \frac{2(h^n - b^{n^2})}{(b^n - h)^2}$$

$$F_1 \equiv \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} + \frac{2a^{n^2}n^s abck}{(h-b^n)^2} + \frac{4a^{n^2}(n^s abck)^2}{(h-b^n)^3} + \frac{8a^{n^2}(n^s abck)^3}{(h-b^n)^4} + \frac{3na^{n(n-1)}(n^s abck)^2 + n^2 a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3}$$

$$\text{ta có: } P = \frac{1}{2} B_1(n^s abck)^3 + \frac{1}{6} C_1(n^s abck)^3 + \frac{1}{2} D(n^s abck)^2 + E(n^s abck) + \frac{h^n - b^{n^2}}{h - b^n} - F_1 =$$

$$= \left[\frac{7nb^{n(n-2)}a^n - 6nh^{n-1} - nh^{n-2}a^n - 6nhb^{n(n-2)}}{2(h-b^n)^3} + \frac{12(h^n - b^{n^2})}{2(h-b^n)^4} \right] (n^s abck)^3 +$$

$$+ \left\{ \frac{[-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1} + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)}}{6(h-b^n)^3} + \frac{12(h^n - b^{n^2})}{6(h-b^n)^4} \right\} (n^s abck)^3 +$$

$$+ \left[\frac{8(h^n - b^{n^2})}{2(h-b^n)^3} - \frac{2nb^{n(n-2)}h + 2nb^{n(n-1)} + 2nh^{n-2}b^n + 2nh^{n-1}}{2(h-b^n)^2} \right] (n^s abck)^2 + \frac{(n^2 - 3n)[h^{n-2} - b^{n(n-2)}]}{2(h-b^n)} (n^s abck)^2 -$$

$$\begin{aligned}
& -\frac{n[h^{n-1} + b^{n(n-1)}]}{h-b^n}(n^s abck) + \frac{2(h^n - b^{n^2})}{(h-b^n)^2}(n^s abck) + \frac{h^n - b^{n^2}}{h-b^n} - \frac{a^{n^2} + na^{n(n-1)}n^s abck}{h-b^n} - \frac{2a^{n^2}n^s abck}{(h-b^n)^2} - \frac{4a^{n^2}(n^s abck)^2}{(h-b^n)^3} - \\
& -\frac{8a^{n^2}(n^s abck)^3}{(h-b^n)^4} - \frac{3na^{n(n-1)}(n^s abck)^2 + n^2a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} \\
\Rightarrow P = & \frac{h^n - b^{n^2} - a^{n^2} - n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}]n^s abck}{h-b^n} + \frac{2(h^n - b^{n^2} - a^{n^2})n^s abck}{(h-b^n)^2} + \frac{4(h^n - b^{n^2} - a^{n^2})}{(h-b^n)^3}(n^s abck)^2 + \\
& + \frac{8(h^n - b^{n^2} - a^{n^2})(n^s abck)^3}{(h-b^n)^4} + \frac{[7nb^{n(n-2)}a^n - 6nh^{n-1} - nh^{n-2}a^n - 6nhb^{n(n-2)}](n^s abck)^3}{2(h-b^n)^3} + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3}(n^s abck)^3 - \\
& - \frac{nb^{n(n-2)}h + nb^{n(n-1)} + nh^{n-2}b^n + nh^{n-1}}{(h-b^n)^2}(n^s abck)^2 + \frac{n^2[h^{n-2} - b^{n(n-2)}](h-b^n)}{2(h-b^n)^2}(n^s abck)^2 - \frac{3n[h^{n-2} - b^{n(n-2)}]}{2(h-b^n)}(n^s abck)^2 - \\
& - \frac{n^2a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} - \frac{3na^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} \quad (13)
\end{aligned}$$

Từ (10), (9), (8), (7) và (13) suy ra:

$$\begin{aligned}
P = P_1 + P_2 = & \frac{n(n-1)}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^2 + \frac{n(n-1)(n-2)}{6}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^3 \\
& + \frac{n(n-1)(n-2)(n-3)}{24}[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4 + 2\frac{n(n-1)(n-2)}{6}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^3 \\
& + \frac{2\left[n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck) + \frac{n(n-1)}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^2\right](n^s abck)}{(b^n - h)^2} - \frac{n^2a^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \\
& + 4\frac{n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck) + \frac{n(n-1)}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^2}{(h-b^n)^3}(n^s abck)^2 + \frac{n^2[h^{n-2} - b^{n(n-2)}](n^s abck)^3}{(h-b^n)^2} + \\
& + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} - \frac{3n}{2}\left(\frac{h^{n-2} - b^{n(n-2)}}{h-b^n}\right)(n^s abck)^2 - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} - \frac{3na^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \\
& + \frac{7nb^{n(n-2)}a^n - 6nh^{n-1} - nh^{n-2}a^n - 6nhb^{n(n-2)}}{2(h-b^n)^3}(n^s abck)^3 - \frac{nb^{n(n-2)}h + nb^{n(n-1)} + nh^{n-2}b^n + nh^{n-1}}{(h-b^n)^2}(n^s abck)^2 + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3}(n^s abck)^3 + \frac{n^2}{2}\left[\frac{h^{n-2} - b^{n(n-2)}}{(h-b^n)^2}\right]a^n(n^s abck)^2 \quad (\text{vì}
\end{aligned}$$

$$h - b^n = a^n + 2n^s abck)$$

với

$$\begin{aligned}
P_1 = & \frac{n(n-1)}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n(n^s abck)^2 + \frac{n(n-1)(n-2)}{6}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3 \\
& + \frac{2n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck) + n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(b^n - h)^2} - \frac{3n[h^{n-2} - b^{n(n-2)}](n^s abck)^2}{2(h-b^n)} +
\end{aligned}$$

$$\begin{aligned}
& + \frac{4n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]}{(h-b^n)^3} (n^s abck)^3 + \frac{[7nb^{n(n-2)}a^n - 6nh^{n-1} - nh^{n-2}a^n - 6nhb^{n(n-2)}](n^s abck)^3}{2(h-b^n)^3} + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 + \frac{n^2[h^{n-2} - b^{n(n-2)}](n^s abck)^3}{(h-b^n)^2} + \\
& - \frac{n[b^{n(n-2)}h + b^{n(n-1)} + h^{n-2}b^n + h^{n-1}]}{2(h-b^n)^2} (n^s abck)^2 + \frac{n^2[h^{n-2} - b^{n(n-2)} - a^{n(n-2)}]a^n (n^s abck)^2}{2(h-b^n)^2} - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} - \frac{3na^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} \\
\text{và } P_2 = & \frac{n(n-1)(n-2)(n-3)[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{24(h-b^n)} + \frac{n(n-1)(n-2)[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(b^n - h)^2} + \\
& + \frac{2n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} \\
\text{. Ta có } P_1 = & \frac{-\frac{n}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n (n^s abck)^2 + \frac{-3n^2 + 2n}{6}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{(h-b^n)^2} + \\
& + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} - \frac{n[b^{n(n-2)}h + b^{n(n-1)} + h^{n-2}b^n + h^{n-1}](n^s abck)^2}{2(h-b^n)^2} + \\
& + \frac{2n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck)^2 + n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(b^n - h)^2} + \frac{4n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck)^3}{(h-b^n)^3} + \\
& + \frac{n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hnb^{n(n-2)}](n^s abck)^3}{2(h-b^n)^3} - \frac{3n[h^{n-2} - b^{n(n-2)}](n^s abck)^2}{2(h-b^n)} + \frac{n^2[h^{n-2} - b^{n(n-2)}](n^s abck)^3}{(h-b^n)^2} + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} - \frac{3na^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} \\
\Rightarrow P_1 = & \frac{-\frac{n}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n (n^s abck)^2 + \frac{-3n^2 + 2n}{6}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{(h-b^n)^2} + \\
& + \frac{2n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck)^2 + n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(b^n - h)^2} + \frac{4n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck)^3}{(h-b^n)^3} + \\
& + \frac{n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hnb^{n(n-2)}](n^s abck)^3}{2(h-b^n)^3} - \frac{n[b^{n(n-2)}h + b^{n(n-1)} + h^{n-2}b^n + h^{n-1}](n^s abck)^2}{(h-b^n)^2} + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 + \frac{n^2[h^{n-2} - b^{n(n-2)}](n^s abck)^3}{(h-b^n)^2} - \\
& - \frac{3n[h^{n-2} - b^{n(n-2)}](n^s abck)^2}{2(h-b^n)} - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} - \frac{3na^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} \\
\Rightarrow P_1 = & \frac{-\frac{n}{2}[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n (n^s abck)^2 - n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3 + \frac{n}{3}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{(h-b^n)^2} + \\
& + \frac{2n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck)^2}{(b^n - h)^2} + \frac{4n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck)^3}{(h-b^n)^3} + \frac{n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hnb^{n(n-2)}](n^s abck)^3}{2(h-b^n)^3} + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 + \frac{n^2[h^{n-2} - b^{n(n-2)}](n^s abck)^3}{(h-b^n)^2} -
\end{aligned}$$

$$\begin{aligned}
& - \frac{n[2b^{n(n-2)}h + 2b^{n(n-1)} + 2h^{n-2}b^n + 2h^{n-1}](n^s abck)^2}{2(h-b^n)^2} - \frac{3n[h^{n-2} - b^{n(n-2)}](n^s abck)^2}{2(h-b^n)} - \frac{3na^{n(n-1)}(n^s abck)^2}{2(h-b^n)^2} \\
& - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \frac{-\frac{n^2}{2}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3 + n^2[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^2} + \\
& + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h-b^n)^2} \\
\Rightarrow P_1 = & \frac{-3n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n(n^s abck)^2 + n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{3(h-b^n)^2} - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \\
& + \frac{n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hb^{n(n-2)}](n^s abck)^3}{2(h-b^n)^3} + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 - \\
& - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h-b^n)^2} + \frac{4n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck)^3}{(h-b^n)^3} + nG(n^s abck)^2 \quad (14) \\
\text{v\o i } G = & \frac{-[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n}{2(h-b^n)^2} + \frac{2[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]}{(b^n - h)^2} - \frac{2b^{n(n-2)}h + 2b^{n(n-1)} + 2h^{n-2}b^n + 2h^{n-1}}{2(h-b^n)^2} - \frac{3[h^{n-2} - b^{n(n-2)}]}{2(h-b^n)} - \frac{3a^{n(n-1)}}{2(h-b^n)^2} \\
= & \frac{-[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n}{2(h-b^n)^2} + \frac{2[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]}{(b^n - h)^2} - \frac{b^{n(n-2)}h + b^{n(n-1)} + h^{n-2}b^n + h^{n-1}}{(h-b^n)^2} - \frac{3[h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^2} (a^n + 2n^s abck) - \\
& - \frac{3a^{n(n-1)}}{2(h-b^n)^2} \text{ (v\i } h - b^n = a^n + 2n^s abck) \\
= & \frac{-[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n + 4[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}] - [2b^{n(n-2)}h + 2b^{n(n-1)} + 2h^{n-2}b^n + 2h^{n-1}] - 3[h^{n-2} - b^{n(n-2)}]a^n - 3a^{n(n-1)}}{2(h-b^n)^2} - \\
& - \frac{3[h^{n-2} - b^{n(n-2)}]n^s abck}{(h-b^n)^2} \\
= & \frac{b^{n(n-1)} + [b^{n(n-2)}a^n - b^{n(n-2)}h] - [h^{n-2}a^n + h^{n-2}b^n] + h^{n-1} - 3[h^{n-2} - b^{n(n-2)}]n^s abck}{(h-b^n)^2} \\
= & \frac{b^{n(n-1)} + b^{n(n-2)}(a^n - h) - h^{n-2}(a^n + b^n) + h^{n-1} - 3[h^{n-2} - b^{n(n-2)}]n^s abck}{(h-b^n)^2} \\
= & \frac{b^{n(n-1)} + b^{n(n-2)}(-b^n - 2n^s abck) - h^{n-2}(h - 2n^s abck) + h^{n-1} - 3[h^{n-2} - b^{n(n-2)}]n^s abck}{(h-b^n)^2} \text{ (v\i } a^n + b^n = h - 2n^s abck, a^n - h = -b^n - 2n^s abck) \\
= & \frac{b^{n(n-1)} - b^{n(n-1)} - 2b^{n(n-2)}n^s abck - h^{n-1} + 2h^{n-2}n^s abck + h^{n-1} - 3[h^{n-2} - b^{n(n-2)}]n^s abck}{(h-b^n)^2} \\
= & \frac{-2b^{n(n-2)}n^s abck + 2h^{n-2}n^s abck - 3[h^{n-2} - b^{n(n-2)}]n^s abck}{(h-b^n)^2} \\
= & \frac{[b^{n(n-2)} - h^{n-2}]n^s abck}{(h-b^n)^2} \quad (15)
\end{aligned}$$

T\ur (14) v\i (15) suy ra: $P_1 = \frac{-n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3 + \frac{n}{3}[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{(h-b^n)^2} - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} +$

$$\begin{aligned}
& + \frac{4n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]}{(h-b^n)^3} (n^s abck)^3 + \frac{n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hb^{n(n-2)}](n^s abck)^3}{2(h-b^n)^3} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 + \frac{n[b^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^2} - \\
& - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} \\
\Rightarrow P_1 = & \frac{-3n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}] + n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n}{3(h-b^n)^2} (n^s abck)^3 + \frac{4n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]}{(h-b^n)^3} (n^s abck)^3 + \\
& + \frac{n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hb^{n(n-2)}]}{2(h-b^n)^3} (n^s abck)^3 - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \frac{n[b^{n(n-2)} - h^{n-2}](h-b^n)(n^s abck)^3}{(h-b^n)^3} + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} - \\
& - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} \\
\Rightarrow P_1 = & \frac{-3n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}] + n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n}{3(h-b^n)^2} (n^s abck)^3 + \frac{4n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]}{(h-b^n)^3} (n^s abck)^3 + \\
& + \frac{n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hb^{n(n-2)}]}{2(h-b^n)^3} (n^s abck)^3 - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} + \frac{2n[b^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \\
& + \frac{[5nb^{n(n-2)} - 2nhb^{n(n-3)}]a^n - 6nb^{n(n-1)} + [-5nh^{n-2} + 2nb^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} - \\
& - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \frac{n[b^{n(n-2)} - h^{n-2}]a^n (n^s abck)^3}{(h-b^n)^3} (\forall h-b^n = a^n + 2n^s abck) \\
\Rightarrow P_1 = & \frac{-6n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n + 2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^{2n}}{6(h-b^n)^3} (n^s abck)^3 + \frac{24n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}](n^s abck)^3}{6(h-b^n)^3} + \\
& + \frac{3n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hb^{n(n-2)}](n^s abck)^3}{6(h-b^n)^3} + \frac{n[5b^{n(n-2)} - 2hb^{n(n-3)}]a^n - 6nb^{n(n-1)} + n[-5h^{n-2} + 2b^n h^{n-3}]a^n - 6nh^{n-1}}{6(h-b^n)^3} (n^s abck)^3 - \\
& - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \frac{n[b^{n(n-2)} - h^{n-2}]a^n (n^s abck)^3}{(h-b^n)^3} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} + \\
& + \frac{2n[b^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} - \frac{14na^{n(n-1)}(n^s abck)^3}{6(h-b^n)^3} \\
\Rightarrow P_1 \equiv & \frac{-6n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n + 2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^{2n} + 24n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]}{6(h-b^n)^3} (n^s abck)^3 + \\
& + \frac{3n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hb^{n(n-2)}] + n[5b^{n(n-2)} - 2hb^{n(n-3)}]a^n - 6nb^{n(n-1)}}{6(h-b^n)^3} (n^s abck)^3 + \\
& + \frac{n[-5h^{n-2} + 2b^n h^{n-3}]a^n - 6nh^{n-1} + 6nb^{n(n-2)}a^n - 6nh^{n-2}a^n - 14na^{n(n-1)}}{6(h-b^n)^3} (n^s abck)^3 - \\
& - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow P_1 &= \frac{-6n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]a^n + 2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^{2n} + 24n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]}{6(h-b^n)^3} (n^s abck)^3 + \\
&+ \frac{3n[7b^{n(n-2)}a^n - 6h^{n-1} - h^{n-2}a^n - 6hb^{n(n-2)}] + n[5b^{n(n-2)} - 2hb^{n(n-3)}]a^n - 6nb^{n(n-1)}}{6(h-b^n)^3} (n^s abck)^3 + \\
&+ \frac{n[-5h^{n-2} + 2b^n h^{n-3}]a^n - 6nh^{n-1} + 6nb^{n(n-2)}a^n - 6nh^{n-2}a^n - 14na^{n(n-1)}}{6(h-b^n)^3} (n^s abck)^3 - \\
&- \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} \\
\Rightarrow P_1 &= \frac{n[(2h^{n-3}a^{2n} + 2b^n h^{n-3}a^n) + (2b^{n(n-3)}a^{2n} - 2hb^{n(n-3)}a^n)] + n[-18hb^{n(n-2)} - 8h^{n-2}a^n + 26b^{n(n-2)}a^n + 18b^{n(n-1)} + 6a^{n(n-1)}]}{6(h-b^n)^3} (n^s abck)^3 - \\
&- \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \frac{n^2[h^{n-2} - b^{n(n-2)}](n^s abck)^3}{(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} \\
\Rightarrow P_1 &= \frac{n[2h^{n-3}a^n(a^n + b^n) - 2b^{n(n-3)}a^n(h - a^n)] + n[-18b^{n(n-2)}(h - a^n) + 8b^{n(n-2)}a^n] - 8nh^{n-2}a^n + 18nb^{n(n-1)} + 6na^{n(n-1)}}{6(h-b^n)^3} (n^s abck)^3 - \\
&- \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} \\
\Rightarrow P_1 &= \frac{n[2h^{n-3}a^n h - 2b^{n(n-3)}a^n b^n] + n[-18b^{n(n-2)}b^n + 8b^{n(n-2)}a^n] - 8nh^{n-2}a^n + 18nb^{n(n-1)} + 6na^{n(n-1)}}{6(h-b^n)^3} (n^s abck)^3 + \\
&+ \frac{n[2h^{n-3}a^n(-2n^s abck) - 2b^{n(n-3)}a^n(2n^s abck)] - n18b^{n(n-2)}(2n^s abck)}{6(h-b^n)^3} (n^s abck)^3 - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \\
&+ \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} \quad (\text{vì } h - a^n = b^n + 2n^s abck, a^n + b^n = h - 2n^s abck) \\
\Rightarrow P_1 &= \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} - \\
&- \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3} (n^s abck)^4,
\end{aligned}$$

Gọi

$$T = P - F_3 = P_1 + P_2 - F_3,$$

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ra

$$\begin{aligned}
T &\equiv \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n (n^s abck)^3}{6(h-b^n)^2} - \\
&- \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3} (n^s abck)^4 + \frac{n(n-1)(n-2)(n-3)[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{24(h-b^n)} + \\
&+ \frac{n(n-1)(n-2)[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(b^n - h)^2} + \frac{2n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \\
&+ \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} - \frac{n^3}{6}a^{n(n-4)}(n^s abck)^3 + \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \pmod{n^{4s+2}}
\end{aligned}$$

$$\text{Ta có } Q_1 = \frac{16(h^n - b^{n^2})}{(h-b^n)^5} (n^s abck)^4 + \frac{1}{2} B_2 (n^s abck)^3 + \frac{1}{6} C_2 (n^s abck)^3 - F_2$$

$$\text{Từ } M(n^s abck)^4 \equiv \frac{16(h^n - b^{n^2})}{(h-b^n)^5} (n^s abck)^4 + M_1(n^s abck)^4 \pmod{n^{4s+2}} \text{ và các biểu thức}$$

$$\begin{aligned}
M_1 &= \frac{7b^{n(n-4)}(b^n-h)^3 + 7b^{n(n-3)}(b^n-h)^2 - 7[-b^{n(n-2)}(b^n-h) - b^{n(n-1)} + h^{n-1}]}{(b^n-h)^4} n + \frac{5[3nb^{n(n-4)}(b^n-h)^2 - 4nb^{n(n-3)}(h-b^n) + 6nb^{n(n-2)}]}{4(h-b^n)^3} \\
&- \frac{3nh^{n-4}(b^n-h)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3} - \frac{nh^{n-1}}{(b^n-h)^4} + \frac{9nb^{n(n-4)}(b^n-h)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3} + \\
&+ \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(b^n-h)^2} - \frac{15nb^{n(n-1)}}{(b^n-h)^4} \\
B_2 &= \frac{2n^2b^{n(n-2)}}{(h-b^n)^2} - \frac{2n^2hb^{n(n-3)}}{(h-b^n)^2} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}]}{(b^n-h)^2}, C_2 = \frac{6n^2h^{n-2} - 3n^2b^nh^{n-3}}{(h-b^n)^2} + \frac{3n^2hb^{n(n-3)} - 6n^2b^{n(n-2)}}{(h-b^n)^2} \\
B_3 &= \frac{14nb^{n(n-2)}n^s abck - 2nh^{n-2}n^s abck}{(h-b^n)^3}, F_2 = \frac{2n^2a^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} + \frac{16a^{n(n-1)}(n^s abck)^4}{(h-b^n)^4} - \frac{n^2}{2}a^{n(n-4)}(n^s abck)^3 \\
C_3 &= \frac{[-5nh^{n-2} + 2nb^nh^{n-3}](2n^s abck) + [5nb^{n(n-2)} - 2nhb^{n(n-3)}](2n^s abck)}{(h-b^n)^3} - \frac{n^3h^{n-2} - n^3b^nh^{n-3}}{(h-b^n)^2} + \frac{n^3b^{n(n-2)} - n^3hb^{n(n-3)}}{(h-b^n)^2} \\
\text{Ta có: } Q &= Q_1 + M_1(n^s abck)^4 + \frac{1}{2}B_3(n^s abck)^3 + \frac{1}{6}C_3(n^s abck)^3 = \frac{16(h^n-b^{n^2})(n^s abck)^4}{(h-b^n)^5} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n-h)^2} + \\
&+ \frac{n^2b^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} + \frac{[2n^2h^{n-2} - n^2b^nh^{n-3}](n^s abck)^3}{2(h-b^n)^2} + \frac{[n^2hb^{n(n-3)} - 2n^2b^{n(n-2)}](n^s abck)^3}{2(h-b^n)^2} - \frac{16a^{n(n-1)}(n^s abck)^4}{(h-b^n)^4} - \\
&- \frac{2n^2a^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} + \frac{n^2a^{n(n-4)}(n^s abck)^3}{2} - \frac{n^2hb^{n(n-3)}(n^s abck)^3}{(h-b^n)^2} + M_1(n^s abck)^4 + \frac{1}{6}C_3(n^s abck)^3 + \frac{1}{2}B_3(n^s abck)^3 \\
&= \frac{16(h^n-b^{n^2})(n^s abck)^4}{(h-b^n)^5} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n-h)^2} + \frac{n^2b^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n^2hb^{n(n-3)}(n^s abck)^3}{(h-b^n)^2} + \\
&+ \frac{[2n^2h^{n-2} - n^2b^nh^{n-3}](n^s abck)^3}{2(h-b^n)^2} + \frac{[n^2hb^{n(n-3)} - 2n^2b^{n(n-2)}](n^s abck)^3}{2(h-b^n)^2} - \frac{16a^{n(n-1)}(n^s abck)^4}{(h-b^n)^4} - \frac{2n^2a^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} + \\
&+ \frac{n^2a^{n(n-4)}(n^s abck)^3}{2} - \frac{n^3h^{n-2} - n^3b^nh^{n-3}}{6(h-b^n)^2}(n^s abck)^3 + \frac{n^3b^{n(n-2)} - n^3hb^{n(n-3)}}{6(h-b^n)^2}(n^s abck)^3 + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h-b^n)^3}(n^s abck)^4 + \\
&+ \frac{[-5nh^{n-2} + 2nb^nh^{n-3}](n^s abck) + [5nb^{n(n-2)} - 2nhb^{n(n-3)}](n^s abck)}{3(h-b^n)^3} - \frac{3nh^{n-4}(b^n-h)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3}(n^s abck)^4 + \\
&+ \frac{7b^{n(n-4)}(b^n-h)^3 + 7b^{n(n-3)}(b^n-h)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4}n(n^s abck)^4 + \frac{5[3nb^{n(n-4)}(b^n-h)^2 - 4nb^{n(n-3)}(h-b^n) + 6b^{n(n-2)}]}{4(h-b^n)^3}(n^s abck)^4 - \\
&- \frac{15nb^{n(n-1)}}{(b^n-h)^4}(n^s abck)^4 - \frac{nh^{n-1}}{(h-b^n)^4}(n^s abck)^4 + \frac{9nb^{n(n-4)}(b^n-h)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3}(n^s abck)^4 + \\
&+ \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h-b^n)^2}(n^s abck)^4
\end{aligned}$$

Ta có $Q + T \equiv 0 \pmod{n^{4s+2}}$ (12) mà $Q + T \equiv R \pmod{n^{4s+2}}$, nên $R \equiv 0 \pmod{n^{4s+2}}$

$$\begin{aligned}
\text{Với } R &= \frac{16(h^n-b^{n^2})(n^s abck)^4}{(h-b^n)^5} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n-h)^2} + \frac{n^2b^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n^2hb^{n(n-3)}(n^s abck)^3}{(h-b^n)^2} + \\
&+ \frac{[2n^2h^{n-2} - n^2b^nh^{n-3}](n^s abck)^3}{2(h-b^n)^2} + \frac{[n^2hb^{n(n-3)} - 2n^2b^{n(n-2)}](n^s abck)^3}{2(h-b^n)^2} - \frac{16a^{n(n-1)}(n^s abck)^4}{(h-b^n)^4} - \frac{2n^2a^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} + \\
&+ \frac{n^2a^{n(n-4)}(n^s abck)^3}{2} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{2(h-b^n)^2} +
\end{aligned}$$

$$\begin{aligned}
& \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h-b^n)^2} - \frac{n^3h^{n-2} - n^3b^nh^{n-3}}{6(h-b^n)^2}(n^s abck)^3 + \frac{n^3b^{n(n-2)} - n^3hb^{n(n-3)}}{6(h-b^n)^2}(n^s abck)^3 + \\
& + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h-b^n)^3}(n^s abck)^4 - \frac{3nh^{n-4}(b^n - h)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3}(n^s abck)^4 + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(b^n - h)^2}(n^s abck)^4 - \\
& + \frac{[-5nh^{n-2} + 2nb^nh^{n-3}] + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]}{3(h-b^n)^3}(n^s abck)^4 + \frac{5[3nb^{n(n-4)}(b^n - h)^2 - 4nb^{n(n-3)}(h-b^n) + 6nb^{n(n-2)}]}{4(h-b^n)^3}(n^s abck)^4 - \\
& - \frac{15nb^{n(n-1)}}{(b^n - h)^4}(n^s abck)^4 + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4}n(n^s abck)^4 - \\
& + \frac{9nb^{n(n-4)}(h-b^n)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3}(n^s abck)^4 + \frac{n}{4}a^{n(n-5)}(n^s abck)^4 + \frac{2na^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} - \\
& - \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3}(n^s abck)^4 - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} - \frac{n^3}{6}a^{n(n-4)}(n^s abck)^3 - \frac{nh^{n-1}}{(h-b^n)^4}(n^s abck)^4 + \\
& + \frac{n(n-1)(n-2)(n-3)[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{24(h-b^n)} + \frac{n(n-1)(n-2)[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(h-b^n)^2} + \\
& + \frac{2n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} \\
& = \frac{16(h^n - b^{n^2})(n^s abck)^4}{(h-b^n)^5} + \frac{n^2b^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n^2hb^{n(n-3)}(n^s abck)^3}{(h-b^n)^2} + \frac{[2n^2h^{n-2} - n^2b^nh^{n-3}](n^s abck)^3}{2(h-b^n)^2} + \\
& + \frac{[n^2hb^{n(n-3)} - 2n^2b^{n(n-2)}](n^s abck)^3}{2(h-b^n)^2} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n - h)^2} + \frac{n^2a^{n(n-4)}(a^n + 2n^s abck)^2(n^s abck)^3}{2(h-b^n)^2} - \\
& - \frac{2n^2a^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{16a^{n(n-1)}(a^n + 2n^s abck)(n^s abck)^4}{(h-b^n)^5} - \frac{n^3h^{n-2} - n^3b^nh^{n-3}}{6(h-b^n)^2}(n^s abck)^3 + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h-b^n)^3}(n^s abck)^4 + \\
& + \frac{n^3b^{n(n-2)} - n^3hb^{n(n-3)}}{6(h-b^n)^2}(n^s abck)^3 - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} - \frac{15nb^{n(n-1)}}{(b^n - h)^4}(n^s abck)^4 - \\
& - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{2(h-b^n)^2} - \frac{3nh^{n-4}(b^n - h)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3}(n^s abck)^4 + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h-b^n)^2}(n^s abck)^4 + \\
& + \frac{5[3nb^{n(n-4)}(b^n - h)^2 - 4nb^{n(n-3)}(h-b^n) + 6nb^{n(n-2)}]}{4(h-b^n)^3}(n^s abck)^4 - \frac{nh^{n-1}}{(h-b^n)^4}(n^s abck)^4 + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h-b^n)^2} + \\
& + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4}n(n^s abck)^4 - \frac{n^3}{6}a^{n(n-4)}(n^s abck)^3 - \\
& + \frac{9nb^{n(n-4)}(h-b^n)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3}(n^s abck)^4 + \frac{2na^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \frac{n}{4}a^{n(n-5)}(n^s abck)^4 + \\
& + \frac{[-5nh^{n-2} + 2nb^nh^{n-3}] + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]}{3(h-b^n)^3}(n^s abck)^4 + \frac{n(n-1)(n-2)[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(b^n - h)^2} - \\
& - \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3}(n^s abck)^4 + \frac{2n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \\
& + \frac{n(n-1)(n-2)(n-3)[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{24(h-b^n)} + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} \text{ (vì } h-b^n = a^n + 2n^s abck)
\end{aligned}$$

$$\begin{aligned}
&= \frac{16(h^n - b^n)(n^s abck)^4}{(h - b^n)^5} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n - h)^2} + \frac{n^2b^{n(n-2)}(n^s abck)^3}{(h - b^n)^2} - \frac{n^2hb^{n(n-3)}(n^s abck)^3}{(h - b^n)^2} + \\
&+ \frac{n^2[2h^{n-2} - b^n h^{n-3}](n^s abck)^3}{2(h - b^n)^2} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{2(h - b^n)^2} - \frac{2n^2a^{n(n-2)}(n^s abck)^3}{(h - b^n)^2} + \frac{n^2a^{n(n-2)}(n^s abck)^3}{2(h - b^n)^2} + \\
&+ \frac{4n^2a^{n(n-4)}(a^n + n^s abck)(n^s abck)^4}{2(h - b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h - b^n)^2} - \frac{16a^{n^2}(n^s abck)^4}{(h - b^n)^5} - \frac{32a^{n(n-1)}(n^s abck)^5}{(h - b^n)^5} + \\
&+ \frac{n^2[hb^{n(n-3)} - 2b^{n(n-2)}](n^s abck)^3}{3(h - b^n)^2} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h - b^n)^3} + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h - b^n)^3}(n^s abck)^4 + \\
&- \frac{n^3h^{n-2} - n^3b^n h^{n-3}}{6(h - b^n)^2}(n^s abck)^3 + \frac{n^3b^{n(n-2)} - n^3hb^{n(n-3)}}{6(h - b^n)^2}(n^s abck)^3 + \frac{5[3nb^{n(n-4)}(h - b^n)^2 - 4nb^{n(n-3)}(h - b^n) + 6nb^{n(n-2)}]}{4(h - b^n)^3}(n^s abck)^4 - \\
&- \frac{15nb^{n(n-1)}}{(b^n - h)^4}(n^s abck)^4 - \frac{3nh^{n-4}(h - b^n)^2 + 4nh^{n-3}(h - b^n) + 6nh^{n-2}}{12(h - b^n)^3}(n^s abck)^4 + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h - b^n)^2}(n^s abck)^4 + \\
&+ \frac{9nb^{n(n-4)}(h - b^n)^2 - 3n[2b^{n(n-3)}(h - b^n) + h^{n-2} - b^{n(n-2)}]}{2(h - b^n)^3}(n^s abck)^4 + \frac{n(n-1)(n-2)[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(b^n - h)^2} \\
&+ \frac{[-5nh^{n-2} + 2nb^n h^{n-3}] + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]}{3(h - b^n)^3}(n^s abck)^4 + \frac{2n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h - b^n)^3} + \\
&+ \frac{-7b^{n(n-4)}(b^n - h)^3 + 7b^{n(n-3)}(h - b^n)^2 - 7[b^{n(n-2)}(h - b^n) + h^{n-1} - b^{n(n-1)}]}{(h - b^n)^4}n(n^s abck)^4 - \frac{nh^{n-1}}{(h - b^n)^4}(n^s abck)^4 + \\
&- \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h - b^n)^3}(n^s abck)^4 - \frac{4na^{n(n-2)}(n^s abck)^4}{(h - b^n)^3} - \frac{n^3a^{n(n-4)}(n^s abck)^3}{6} + \frac{2na^{n(n-3)}(n^s abck)^4}{(h - b^n)^2} + \\
&+ \frac{n(n-1)(n-2)(n-3)[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{24(h - b^n)} + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h - b^n)^4} + \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \\
&\Rightarrow \frac{16(h^n - b^n)(n^s abck)^4}{(h - b^n)^5} + \frac{n^2b^{n(n-2)}(n^s abck)^3}{(h - b^n)^2} - \frac{n^2hb^{n(n-3)}(n^s abck)^3}{(h - b^n)^2} + \frac{[2n^2h^{n-2} - n^2b^n h^{n-3}](n^s abck)^3}{2(h - b^n)^2} + \\
&+ \frac{[n^2hb^{n(n-3)} - 2n^2b^{n(n-2)}](n^s abck)^3}{2(h - b^n)^2} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n - h)^2} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h - b^n)^3} + \\
&+ \frac{n^2a^{n(n-2)}(n^s abck)^3}{2(h - b^n)^2} - \frac{2n^2a^{n(n-2)}(n^s abck)^3}{(h - b^n)^2} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{2(h - b^n)^2} - \frac{16a^{n^2}(n^s abck)^4}{(h - b^n)^5} + \\
&+ \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h - b^n)^2} - \frac{n^3h^{n-2} - n^3b^n h^{n-3}}{6(h - b^n)^2}(n^s abck)^3 + \frac{n^3b^{n(n-2)} - n^3hb^{n(n-3)}}{6(h - b^n)^2}(n^s abck)^3 + \\
&+ \frac{7nb^{n(n-2)} - nh^{n-2}}{(h - b^n)^3}(n^s abck)^4 + \frac{5[3nb^{n(n-4)}(b^n - h)^2 - 4nb^{n(n-3)}(h - b^n) + 6nb^{n(n-2)}]}{4(h - b^n)^3}(n^s abck)^4 - \\
&- \frac{15nb^{n(n-1)}}{(b^n - h)^4}(n^s abck)^4 - \frac{3nh^{n-4}(b^n - h)^2 + 4nh^{n-3}(h - b^n) + 6nh^{n-2}}{12(h - b^n)^3}(n^s abck)^4 + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h - b^n)^2}(n^s abck)^4 + \\
&+ \frac{-7b^{n(n-4)}(h - b^n)^3 + 7b^{n(n-3)}(h - b^n)^2 - 7[b^{n(n-2)}(h - b^n) + h^{n-1} - b^{n(n-1)}]}{(h - b^n)^4}n(n^s abck)^4 - \frac{nh^{n-1}}{(h - b^n)^4}(n^s abck)^4 - \\
&+ \frac{9nb^{n(n-4)}(h - b^n)^2 - 3n[2b^{n(n-3)}(h - b^n) + h^{n-2} - b^{n(n-2)}]}{2(h - b^n)^3}(n^s abck)^4 + \frac{[-5nh^{n-2} + 2nb^n h^{n-3}] + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]}{3(h - b^n)^3}(n^s abck)^4 -
\end{aligned}$$

$$\begin{aligned}
& -\frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3}(n^s abck)^4 + \frac{2n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \\
& + \frac{n(n-1)(n-2)(n-3)[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{24(h-b^n)} - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} - \frac{n^3}{6}a^{n(n-4)}(n^s abck)^3 + \frac{2na^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \\
& + \frac{n}{4}a^{n(n-5)}(n^s abck)^4 + \frac{2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(b^n - h)^2} + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} \pmod{n^{4s+2}} \\
\Rightarrow & \frac{16(h^n - b^{n^2})(n^s abck)^4}{(h-b^n)^5} + \frac{n^2b^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n^2hb^{n(n-3)}(n^s abck)^3}{(h-b^n)^2} + \frac{[2n^2h^{n-2} - n^2b^nh^{n-3}](n^s abck)^3}{2(h-b^n)^2} + \\
& + \frac{[n^2hb^{n(n-3)} - 2n^2b^{n(n-2)}](n^s abck)^3}{2(h-b^n)^2} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n - h)^2} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} + \\
& + \frac{n^2a^{n(n-2)}(n^s abck)^3}{2(h-b^n)^2} - \frac{2n^2a^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{2(h-b^n)^2} - \frac{16a^{n^2}(n^s abck)^4}{(h-b^n)^5} + \\
& - \frac{n^3h^{n-2} - n^3b^nh^{n-3}}{6(h-b^n)^2}(n^s abck)^3 + \frac{n^3b^{n(n-2)} - n^3hb^{n(n-3)}}{6(h-b^n)^2}(n^s abck)^3 + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h-b^n)^3}(n^s abck)^4 + \\
& + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h-b^n)^2} + \frac{5[3nb^{n(n-4)}(h-b^n)^2 - 4nb^{n(n-3)}(h-b^n) + 6nb^{n(n-2)}]}{4(h-b^n)^3}(n^s abck)^4 - \frac{15nb^{n(n-1)}}{(h-b^n)^4}(n^s abck)^4 - \\
& - \frac{3nh^{n-4}(h-b^n)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3}(n^s abck)^4 + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4}n(n^s abck)^4 - \\
& + \frac{9nb^{n(n-4)}(h-b^n)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3}(n^s abck)^4 - \frac{nh^{n-1}}{(h-b^n)^4}(n^s abck)^4 + \\
& + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h-b^n)^2}(n^s abck)^4 + \frac{[-5nh^{n-2} + 2nb^nh^{n-3}] + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]}{3(h-b^n)^3}(n^s abck)^4 - \\
& - \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3}(n^s abck)^4 + \frac{2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(h-b^n)^2} + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} + \\
& + \frac{n(n-1)(n-2)(n-3)[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{24(h-b^n)} + \frac{2(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} - \\
& - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} - \frac{n^3}{6}a^{n(n-4)}(n^s abck)^3 + \frac{2na^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \pmod{n^{4s+2}} \\
\Rightarrow & \frac{16(h^n - b^{n^2} - a^{n^2})(n^s abck)^4}{(h-b^n)^5} + \frac{n^2b^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n^2hb^{n(n-3)}(n^s abck)^3}{(h-b^n)^2} + \frac{[2n^2h^{n-2} - n^2b^nh^{n-3}](n^s abck)^3}{2(h-b^n)^2} + \\
& + \frac{[n^2hb^{n(n-3)} - 2n^2b^{n(n-2)}](n^s abck)^3}{2(h-b^n)^2} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n - h)^2} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} + \\
& - \frac{3n^2a^{n(n-2)}(n^s abck)^3}{2(h-b^n)^2} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{2(h-b^n)^2} + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h-b^n)^3}(n^s abck)^4 + \\
& - \frac{n^3h^{n-2} - n^3b^nh^{n-3}}{6(h-b^n)^2}(n^s abck)^3 + \frac{n^3b^{n(n-2)} - n^3hb^{n(n-3)}}{6(h-b^n)^2}(n^s abck)^3 + \frac{5[3nb^{n(n-4)}(b^n - h)^2 - 4nb^{n(n-3)}(h-b^n) + 6nb^{n(n-2)}]}{4(h-b^n)^3}(n^s abck)^4 + \\
& + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h-b^n)^2} - \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3}(n^s abck)^4 +
\end{aligned}$$

$$\begin{aligned}
& -\frac{15nb^{n(n-1)}}{(b^n-h)^4}(n^s abck)^4 - \frac{3nh^{n-4}(h-b^n)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3}(n^s abck)^4 + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h-b^n)^2}(n^s abck)^4 + \\
& + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4}n(n^s abck)^4 - \frac{nh^{n-1}}{(h-b^n)^4}(n^s abck)^4 - \\
& + \frac{9nb^{n(n-4)}(h-b^n)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3}(n^s abck)^4 - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} - \frac{n^3}{6}a^{n(n-4)}(n^s abck)^3 + \\
& + \frac{[-5nh^{n-2} + 2nb^n h^{n-3}] + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]}{3(h-b^n)^3}(n^s abck)^4 + \frac{2n(n-1)[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \\
& + \frac{n(n-1)(n-2)(n-3)[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{24(h-b^n)} + \frac{n}{4}a^{n(n-5)}(n^s abck)^4 + \frac{2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(h-b^n)^2} + \\
& + \frac{2na^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} \equiv 0 \pmod{n^{4s+2}} \\
\Rightarrow & \frac{16n(h^{n-1} + a^{n(n-1)} + b^{n(n-1)})(n^s abck)^5}{(h-b^n)^5} + \frac{n^2 b^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n^2 hb^{n(n-3)}(n^s abck)^3}{(h-b^n)^2} + \frac{[2n^2 h^{n-2} - n^2 b^n h^{n-3}](n^s abck)^3}{2(h-b^n)^2} + \\
& + \frac{[n^2 hb^{n(n-3)} - 2n^2 b^{n(n-2)}](n^s abck)^3}{2(h-b^n)^2} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(h-b^n)^2} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} - \\
& - \frac{3n^2 a^{n(n-2)}(n^s abck)^3}{2(h-b^n)^2} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}][a^n(n^s abck)^3]}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}][a^n(n^s abck)^3]}{6(h-b^n)^2} - \\
& - \frac{n^3 h^{n-2} - n^3 b^n h^{n-3}}{6(h-b^n)^2}(n^s abck)^3 + \frac{n^3 b^{n(n-2)} - n^3 hb^{n(n-3)}}{6(h-b^n)^2}(n^s abck)^3 + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h-b^n)^3}(n^s abck)^4 + \frac{2na^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \\
& + \frac{5[3nb^{n(n-4)}(h-b^n)^2 - 4nb^{n(n-3)}(h-b^n) + 6nb^{n(n-2)}]}{4(h-b^n)^3}(n^s abck)^4 - \frac{15nb^{n(n-1)}}{(h-b^n)^4}(n^s abck)^4 - \frac{3nh^{n-4}(h-b^n)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3}(n^s abck)^4 + \\
& + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4}n(n^s abck)^4 - \frac{nh^{n-1}}{(h-b^n)^4}(n^s abck)^4 - \\
& + \frac{9nb^{n(n-4)}(h-b^n)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3}(n^s abck)^4 + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h-b^n)^2}(n^s abck)^4 + \\
& + \frac{[-5nh^{n-2} + 2nb^n h^{n-3}] + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]}{3(h-b^n)^3}(n^s abck)^4 - \frac{n^3}{6}a^{n(n-4)}(n^s abck)^3 + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} - \\
& - \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3}(n^s abck)^4 - \frac{2n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \frac{n}{4}a^{n(n-5)}(n^s abck)^4 \\
& - \frac{n[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{4(h-b^n)} - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} + \frac{2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(b^n-h)^2} \equiv 0 \pmod{n^{4s+2}} \text{ (vì)}
\end{aligned}$$

$$h^n - a^{n^2} - b^{n^2} \equiv n[a^{n(n-1)} + b^{n(n-1)} + h^{n-1}]n^s abck \pmod{n^{2s+1}} \quad (10)$$

$$\begin{aligned}
\Rightarrow & \frac{n^2 b^{n(n-2)}(n^s abck)^3}{(h-b^n)^2} - \frac{n^2 hb^{n(n-3)}(n^s abck)^3}{(h-b^n)^2} + \frac{[2n^2 h^{n-2} - n^2 b^n h^{n-3}](n^s abck)^3}{2(h-b^n)^2} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} + \\
& + \frac{[n^2 hb^{n(n-3)} - 2n^2 b^{n(n-2)}](n^s abck)^3}{2(h-b^n)^2} + \frac{n^2[-3b^{n(n-2)} + 2hb^{n(n-3)} + h^{n-2}](n^s abck)^3}{2(b^n-h)^2} - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}][a^n(n^s abck)^3]}{2(h-b^n)^2} - \\
& - \frac{3n^2 a^{n(n-2)}(n^s abck)^3}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}][a^n(n^s abck)^3]}{6(h-b^n)^2} - \frac{n^3 h^{n-2} - n^3 b^n h^{n-3}}{6(h-b^n)^2}(n^s abck)^3 + \frac{n^3 b^{n(n-2)} - n^3 hb^{n(n-3)}}{6(h-b^n)^2}(n^s abck)^3 +
\end{aligned}$$

$$\begin{aligned}
& + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h-b^n)^3} (n^s abck)^4 + \frac{5[3nb^{n(n-4)}(h-b^n)^2 - 4nb^{n(n-3)}(h-b^n) + 6nb^{n(n-2)}]}{4(h-b^n)^3} (n^s abck)^4 - \frac{15nb^{n(n-1)}}{(h-b^n)^4} (n^s abck)^4 - \frac{nh^{n-1}}{(h-b^n)^4} (n^s abck)^4 - \\
& - \frac{3nh^{n-4}(h-b^n)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3} (n^s abck)^4 + \frac{9nb^{n(n-4)}(h-b^n)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3} (n^s abck)^4 - \\
& - \frac{2n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(b^n - h)^2} (n^s abck)^4 + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} + \\
& - \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3} (n^s abck)^4 + \frac{[-5nh^{n-2} + 2nb^n h^{n-3}] + [5nb^{n(n-2)} - 2nhb^{n(n-3)}]}{3(h-b^n)^3} (n^s abck)^4 - \\
& - \frac{n[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{4(h-b^n)} - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} - \frac{n^3}{6} a^{n(n-4)}(n^s abck)^3 + \frac{2na^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \\
& + \frac{2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(b^n - h)^2} + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4} n(n^s abck)^4 + \\
& + \frac{n}{4} a^{n(n-5)}(n^s abck)^4 \equiv 0 \pmod{n^{4s+2}} \\
\Rightarrow & \frac{n^2 \{-3[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}] + b^n h[b^{n(n-4)} - h^{n-4}]\}(n^s abck)^3}{2(h-b^n)^2} + \frac{na^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^3}{(h-b^n)^3} - \frac{n^3}{6} a^{n(n-4)}(n^s abck)^3 - \\
& - \frac{n^2[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{2(h-b^n)^2} + \frac{n^3[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n(n^s abck)^3}{6(h-b^n)^2} - \frac{n^3 h^{n-2} - n^3 b^n h^{n-3}}{6(h-b^n)^2} (n^s abck)^3 + \\
& + \frac{n^3 b^{n(n-2)} - n^3 h b^{n(n-3)}}{6(h-b^n)^2} (n^s abck)^3 + \frac{7nb^{n(n-2)} - nh^{n-2}}{(h-b^n)^3} (n^s abck)^4 - \frac{15nb^{n(n-1)}}{(h-b^n)^4} (n^s abck)^4 - \frac{nh^{n-1}}{(h-b^n)^4} (n^s abck)^4 + \\
& + \frac{5[3nb^{n(n-4)}(h-b^n)^2 - 4nb^{n(n-3)}(h-b^n) + 6nb^{n(n-2)}]}{4(h-b^n)^3} (n^s abck)^4 - \frac{3nh^{n-4}(h-b^n)^2 + 4nh^{n-3}(h-b^n) + 6nh^{n-2}}{12(h-b^n)^3} (n^s abck)^4 + \\
& + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4} n(n^s abck)^4 + \frac{4nb^{n(n-3)} - 3nhb^{n(n-4)} - nh^{n-3}}{3(h-b^n)^2} (n^s abck)^4 + \\
& + \frac{9nb^{n(n-4)}(h-b^n)^2 - 3n[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3} (n^s abck)^4 + \frac{[-5h^{n-2} + 2b^n h^{n-3}] + [5b^{n(n-2)} - 2hb^{n(n-3)}]}{3(h-b^n)^3} n(n^s abck)^4 - \\
& - \frac{2nh^{n-3}a^n + 2nb^{n(n-3)}a^n + 18nb^{n(n-2)}}{3(h-b^n)^3} (n^s abck)^4 - \frac{2n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)^4}{(h-b^n)^3} - \frac{n[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)^4}{4(h-b^n)} + \\
& + \frac{n}{4} a^{n(n-5)}(n^s abck)^4 - \frac{4na^{n(n-2)}(n^s abck)^4}{(h-b^n)^3} + \frac{2na^{n(n-3)}(n^s abck)^4}{(h-b^n)^2} + \frac{2n[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)^4}{3(h-b^n)^2} + \\
& + \frac{8n[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)^4}{(h-b^n)^4} \equiv 0 \pmod{n^{4s+2}} \\
\Rightarrow & \frac{n\{-3[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}] + b^n h[b^{n(n-4)} - h^{n-4}]\}}{2(h-b^n)^2} + \frac{a^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} - \frac{h^{n-2} - b^{n(n-2)} - b^n h[h^{n-4} - b^{n(n-4)}]}{6(h-b^n)^2} n^2 + \\
& - \frac{n[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n}{2(h-b^n)^2} + \frac{n^2[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n}{6(h-b^n)^2} - \frac{n^2}{6} a^{n(n-4)} + \frac{7b^{n(n-2)} - h^{n-2}}{(h-b^n)^3} (n^s abck) - \frac{15b^{n(n-1)}}{(h-b^n)^4} n^s abck + \\
& + \frac{5[3b^{n(n-4)}(b^n - h)^2 - 4b^{n(n-3)}(h-b^n) + 6b^{n(n-2)}]}{4(h-b^n)^3} (n^s abck) - \frac{3h^{n-4}(b^n - h)^2 + 4h^{n-3}(h-b^n) + 6h^{n-2}}{12(h-b^n)^3} (n^s abck) - \frac{h^{n-1}}{(b^n - h)^4} n^s abck + \frac{1}{4} a^{n(n-5)}(n^s abck) +
\end{aligned}$$

$$\begin{aligned}
& + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4} (n^s abck) + \frac{4b^{n(n-3)} - 3hb^{n(n-4)} - h^{n-3}}{3(h-b^n)^2} (n^s abck) - \\
& + \frac{9b^{n(n-4)}(h-b^n)^2 - 3[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3} (n^s abck) + \frac{[-5h^{n-2} + 2b^n h^{n-3}] + [5b^{n(n-2)} - 2hb^{n(n-3)}]}{3(h-b^n)^3} (n^s abck) - \\
& - \frac{2h^{n-3}a^n + 2b^{n(n-3)}a^n + 18b^{n(n-2)}}{3(h-b^n)^3} (n^s abck) - \frac{[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)}{4(h-b^n)} - \frac{n^2}{6} a^{n(n-4)} + \frac{2a^{n(n-3)}(n^s abck)}{(h-b^n)^2} - \frac{4a^{n(n-2)}(n^s abck)}{(h-b^n)^3} + \\
& - \frac{2[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)}{(h-b^n)^3} + \frac{2[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)}{3(b^n - h)^2} + \frac{8[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)}{(h-b^n)^4} \equiv 0 \pmod{n^{s+1}} \quad (16')
\end{aligned}$$

Từ (16') suy ra: $b^{n(n-2)} + a^{n(n-2)} - h^{n-2} \equiv 0 \pmod{n}$ Đặt:

$$\begin{aligned}
H(h, b) = & \frac{n\{-3[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}] + b^n h[b^{n(n-4)} - h^{n-4}]\}}{2(h-b^n)^2} + \frac{a^n [b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} - \frac{h^{n-2} - b^{n(n-2)} - b^n h[h^{n-4} - b^{n(n-4)}]}{6(h-b^n)^2} n^2 - \\
& - \frac{n[b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n}{2(h-b^n)^2} + \frac{n^2 [b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n}{6(h-b^n)^2} - \frac{n^2}{6} a^{n(n-4)}
\end{aligned}$$

Khi đó (16') trở thành

$$\begin{aligned}
H(h, b) + & \frac{7b^{n(n-2)} - h^{n-2}}{(h-b^n)^3} (n^s abck) - \frac{15b^{n(n-1)}}{(h-b^n)^4} n^s abck + \frac{5[3b^{n(n-4)}(b^n - h)^2 - 4b^{n(n-3)}(h-b^n) + 6b^{n(n-2)}]}{4(h-b^n)^3} (n^s abck) \\
& - \frac{3h^{n-4}(b^n - h)^2 + 4h^{n-3}(h-b^n) + 6h^{n-2}}{12(h-b^n)^3} (n^s abck) - \frac{h^{n-1}}{(b^n - h)^4} n^s abck + \frac{4b^{n(n-3)} - 3hb^{n(n-4)} - h^{n-3}}{3(h-b^n)^2} (n^s abck) - \frac{4a^{n(n-2)}(n^s abck)}{(h-b^n)^3} + \\
& + \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) + h^{n-1} - b^{n(n-1)}]}{(h-b^n)^4} n^s abck - \frac{2[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}](n^s abck)}{(h-b^n)^3} + \\
& + \frac{9b^{n(n-4)}(h-b^n)^2 - 3[2b^{n(n-3)}(h-b^n) + h^{n-2} - b^{n(n-2)}]}{2(h-b^n)^3} (n^s abck) + \frac{[-5h^{n-2} + 2b^n h^{n-3}] + [5b^{n(n-2)} - 2hb^{n(n-3)}]}{3(h-b^n)^3} (n^s abck) - \\
& - \frac{2h^{n-3}a^n + 2b^{n(n-3)}a^n + 18b^{n(n-2)}}{3(h-b^n)^3} (n^s abck) - \frac{[b^{n(n-4)} + a^{n(n-4)} - h^{n-4}](n^s abck)}{4(h-b^n)} + \frac{2a^{n(n-3)}(n^s abck)}{(h-b^n)^2} - \\
& + \frac{2[b^{n(n-3)} + a^{n(n-3)} + h^{n-3}](n^s abck)}{3(b^n - h)^2} + \frac{1}{4} a^{n(n-5)} (n^s abck) + \frac{8[h^{n-1} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)}{(h-b^n)^4} \equiv 0 \pmod{n^{s+1}} \quad (16'')
\end{aligned}$$

1.1 Tính chất của H(h, b) và H(h, a)

$$\begin{aligned}
\text{Ta có } H(h, b) = & \frac{n\{-3[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}] + b^n h[b^{n(n-4)} - h^{n-4}] - [b^{n(n-3)} - a^{n(n-3)} + h^{n-3}]a^n\}}{2(h-b^n)^2} \\
& - \frac{h^{n-2} - b^{n(n-2)} - b^n h[h^{n-4} - b^{n(n-4)}]}{6(h-b^n)^2} n^2 + \frac{n^2 [b^{n(n-3)} + a^{n(n-3)} + h^{n-3}]a^n}{6(h-b^n)^2} - \frac{n^2}{6} a^{n(n-4)} + \frac{a^n [b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} - \\
& = \frac{n\{-3[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}] + [hb^{n(n-3)} - b^{n(n-3)}a^n] - [b^n h^{n-3} + h^{n-3}a^n] + a^{n(n-2)}\}}{2(h-b^n)^2} + \\
& + \frac{b^{n(n-3)}a^n + a^{n(n-2)} + h^{n-3}a^n - h^{n-3}(h-b^n) - b^{n(n-3)}(h-b^n) - a^{n(n-4)}(h-b^n)^2}{6(h-b^n)^2} n^2 + \frac{a^n [b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} \\
& = \frac{n\{-3[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}] - b^{n(n-3)}(h-a^n) - h^{n-3}(a^n + b^n) + a^{n(n-2)}\}}{2(h-b^n)^2} + \frac{a^n [b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} +
\end{aligned}$$

$$\begin{aligned}
& + \frac{b^{n(n-3)}a^n + a^{n(n-2)} + h^{n-3}a^n - h^{n-3}(h-b^n) - b^{n(n-3)}(h-b^n) - a^{n(n-4)}(h-b^n)^2}{6(h-b^n)^2} n^2 \text{ (vì } h = a^n + b^n + 2n^s abck \text{)} \\
& = \frac{n\{-3[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}] - b^{n(n-3)}(b^n + 2n^s abck) - h^{n-3}(h - 2n^s abck) + a^{n(n-2)}\}}{2(h-b^n)^2} + \frac{a^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} \\
& + \frac{b^{n(n-3)}a^n + a^{n(n-2)} + h^{n-3}a^n - h^{n-3}(a^n + 2n^s abck) - b^{n(n-3)}(a^n + 2n^s abck) - a^{n(n-4)}(a^n + 2n^s abck)^2}{6(h-b^n)^2} n^2 \\
& \Rightarrow H(h, b) \equiv \frac{n\{-3[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}] - b^{n(n-3)} - h^{n-2} + a^{n(n-2)}\}}{2(h-b^n)^2} + \frac{a^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} \\
& + \frac{b^{n(n-3)}a^n + a^{n(n-2)} + h^{n-3}a^n - h^{n-3}a^n - b^{n(n-3)}a^n - a^{n(n-4)}a^{2n}}{6(h-b^n)^2} n^2 \pmod{n^{s+1}} \\
& \Rightarrow H(h, b) \equiv \frac{-n[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} (h-b^n) + \frac{a^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(c^n - b^n)^3} \pmod{n^{s+1}} \\
& \Rightarrow H(h, b) \equiv \frac{-n[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} (a^n + 2n^s abck) + \frac{a^n[b^{n(n-2)} + a^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} \pmod{n^{s+1}} \\
& \Rightarrow H(h, b) \equiv \frac{(1-n)a^n[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} \pmod{n^{s+1}} \quad (17)
\end{aligned}$$

Do a, b tính đối xứng trong công thức nghiệm của PT(1) (xem bổ đề 6) nên ta cũng có

$$H(h, a) \equiv \frac{(1-n)b^n[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}]}{(h-a^n)^3} \pmod{n^{s+1}} \quad (17')$$

Vậy ta có $H(h, b) \equiv \frac{(1-n)a^n[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} \pmod{n^{s+1}} \quad (17)$, $H(h, a) \equiv \frac{(1-n)b^n[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}]}{(h-a^n)^3} \pmod{n^{s+1}} \quad (17')$.

1.2. Nếu $h = c^n$

Vì n là số nguyên tố lớn hơn 5, $abc \nmid n$, $a^n + b^n - c^n \equiv 0 \pmod{n^s}$, $s \geq 2$ và $a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv 0 \pmod{n}$ nên theo bổ đề 7, ta có $c^n b^n + a^{2n} \equiv c^n a^n + b^{2n} \equiv b^n a^n - c^{2n} \equiv 0 \pmod{n^2} \quad (18)$, $c^{3n} \equiv -b^{3n} \pmod{n^2} \quad (19)$, $a^{3n} \equiv b^{3n} \pmod{n^2} \quad (20)$, $a^{n(n-4)} + c^{n(n-4)} \equiv b^{n(n-4)} + c^{n(n-4)} \equiv a^{n(n-4)} - b^{n(n-4)} \equiv 0 \pmod{n^2} \quad (21)$, $a^{n(n-3)} + b^{n(n-3)} + c^{n(n-3)} \equiv 0 \pmod{n^2} \quad (22)$, $a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv 0 \pmod{n^2} \quad (22')$ và $n \equiv 1 \pmod{6}$. Từ (17), (21), (22), ta

$$\begin{aligned}
& \text{có: } H(c^n, b) + \frac{7b^{n(n-2)} - c^{n(n-2)}}{(c^n - b^n)^4} (c^n - b^n)(n^s abck) - \frac{2a^n[c^{n(n-3)} + b^{n(n-3)}](c^n - b^n)18b^{n(n-2)}(c^n - b^n)}{3(c^n - b^n)^4} (n^s abck) - \\
& - \frac{3c^{n(n-4)}(c^n - b^n)^3 + 4b^{n(n-3)}(c^n - b^n)^2 + 6c^{n(n-2)}(c^n - b^n)}{12(c^n - b^n)^4} (n^s abck) + \frac{-4b^{n(n-4)}(c^n - b^n)^3 - c^n[c^{n(n-4)} - b^{n(n-4)}](c^n - b^n)^2}{3(c^n - b^n)^4} (n^s abck) + \\
& + \frac{-7b^{n(n-4)}(c^n - b^n)^3 + 7b^{n(n-3)}(c^n - b^n)^2 - 7[b^{n(n-2)}(c^n - b^n) + c^{n(n-1)} - b^{n(n-1)}]}{(c^n - b^n)^4} (n^s abck) - \frac{c^{n(n-1)}}{(c^n - b^n)^4} n^s abck + \\
& + \frac{-5[c^{n(n-2)} - b^{n(n-2)}](c^n - b^n) + 2b^n c^n [c^{n(n-4)} - b^{n(n-4)}](c^n - b^n)}{3(c^n - b^n)^4} (n^s abck) - \frac{a^{n(n-5)}(c^n - b^n)^4 (n^s abck)}{4(c^n - b^n)^4} \\
& + \frac{9b^{n(n-4)}(c^n - b^n)^3 - 3[2b^{n(n-3)}(c^n - b^n)^2 + [c^{n(n-2)} - b^{n(n-2)}](c^n - b^n)]}{2(c^n - b^n)^4} (n^s abck) - \frac{15b^{n(n-1)}}{(c^n - b^n)^4} n^s abck \\
& - \frac{[b^{n(n-4)} + a^{n(n-4)} - c^{n(n-4)}](c^n - b^n)^3 (n^s abck)}{4(c^n - b^n)^4} - \frac{4a^{n(n-2)}(c^n - b^n)(n^s abck)}{(c^n - b^n)^4} + \frac{8[c^{n(n-1)} + b^{n(n-1)} + a^{n(n-1)}](n^s abck)}{(c^n - b^n)^4} + \\
& + \frac{5[3b^{n(n-4)}(c^n - b^n)^3 - 4b^{n(n-3)}(c^n - b^n)^2 + 6b^{n(n-2)}(c^n - b^n)]}{4(c^n - b^n)^4} (n^s abck) + \frac{2a^{n(n-3)}(c^n - b^n)^2 (n^s abck)}{(c^n - b^n)^4} \equiv 0 \pmod{n^{s+1}}
\end{aligned}$$

$$\begin{aligned}
(\text{vì}) \quad & h^{n-4} = c^{n(n-4)} \equiv -a^{n(n-4)} \pmod{n^2}, b^{n(n-4)} \equiv a^{n(n-4)} \pmod{n^2}, c^n a^n \equiv -b^{2n} \pmod{n^2}, c^n b^n \equiv a^{2n} \pmod{n^2} \\
& -b^{n(n-1)} + c^{n(n-1)} = [-b^{n(n-1)} + 1] + [c^{n(n-1)} - 1] \equiv 0 \pmod{n^2}, a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv a^{n(n-3)} + b^{n(n-3)} + c^{n(n-3)} \equiv 0 \pmod{n^2}) \\
\Rightarrow & H(c^n, b) + \{7b^{n(n-2)} a^n - [a^{n(n-2)} - b^{n(n-2)}] a^n\} (n^s abck) - \frac{2(-a^{n(n-3)}) a^n + 18b^{n(n-2)} a^n}{3} (n^s abck) + \\
& + \frac{5[3a^{n(n-4)} a^{3n} - 4b^{n(n-3)} a^{2n} + 6b^{n(n-2)} a^n]}{4} (n^s abck) - \frac{3(-a^{n(n-4)}) a^{3n} + 4b^{n(n-3)} a^{2n} + 6c^{n(n-2)} a^n}{12} (n^s abck) + \frac{-5a^{n(n-1)} - 4a^{n(n-1)}}{3} (n^s abck) + \\
& + [-7a^{n(n-4)} a^{3n} + 7b^{n(n-3)} a^{2n} - 7b^{n(n-2)} a^n] (n^s abck) - a^{n(n-1)} n^s abck - \frac{4a^{n(n-4)} a^{3n} + c^n [-a^{n(n-4)} - a^{n(n-4)}] a^{2n}}{3} (n^s abck) + \\
& + \frac{9a^{n(n-4)} a^{3n} - 3[2b^{n(n-3)} a^{2n} + a^{n(n-2)} a^n]}{2} (n^s abck) - 15a^{n(n-1)} n^s abck - 4a^{n(n-2)} a^n (n^s abck) + \frac{a^{n(n-5)} a^{4n} (n^s abck)}{4} - \\
& - \frac{[a^{n(n-4)} + a^{n(n-4)} - (-a^{n(n-4)})] a^{3n} (n^s abck)}{4} + 2a^{n(n-3)} a^{2n} (n^s abck) + 8[a^{n(n-1)} + a^{n(n-1)} + a^{n(n-1)}] (n^s abck) \equiv 0 \pmod{n^{s+1}}
\end{aligned}$$

$$\begin{aligned}
(\text{vì}) \quad & c^{n(n-4)} \equiv -a^{n(n-4)} \pmod{n^2}, b^{n(n-4)} \equiv a^{n(n-4)} \pmod{n^2}, c^n b^n \equiv a^{2n} \pmod{n^2}, a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv 0 \pmod{n^2}, \\
& b^{n(n-1)} \equiv c^{n(n-1)} \equiv a^{n(n-1)} \pmod{n^2})
\end{aligned}$$

$$\begin{aligned}
\Rightarrow & H(c^n, b) + [6b^{n(n-2)} a^n - a^{n(n-1)}] (n^s abck) - \frac{-2a^{n(n-1)} + 18b^{n(n-2)} a^n}{3} (n^s abck) - \frac{4a^{n(n-1)} - 2c^n a^{n(n-2)}}{3} (n^s abck) + \\
& + \frac{5[3a^{n(n-1)} - 4b^{n(n-3)} a^{2n} + 6b^{n(n-2)} a^n]}{4} (n^s abck) - \frac{-3a^{n(n-1)} + 4b^{n(n-3)} a^{2n} + 6c^{n(n-2)} a^n}{12} (n^s abck) + \frac{-5a^{n(n-1)} - 4a^{n(n-1)}}{3} (n^s abck) - a^{n(n-1)} n^s abck + \\
& + [-7a^{n(n-1)} + 7b^{n(n-3)} a^{2n} - 7b^{n(n-2)} a^n] (n^s abck) + \frac{9a^{n(n-1)} - 3[2b^{n(n-3)} a^{2n} + a^{n(n-1)}]}{2} (n^s abck) - 15a^{n(n-1)} n^s abck - \\
& - \frac{3a^{n(n-1)} (n^s abck)}{4} + 2a^{n(n-1)} (n^s abck) - 4a^{n(n-1)} (n^s abck) + \frac{a^{n(n-1)} (n^s abck)}{4} + 24a^{n(n-1)} (n^s abck) \equiv 0 \pmod{n^{s+1}} \\
\Rightarrow & H(c^n, b) + \frac{2c^n a^{n(n-2)}}{3} (n^s abck) - \frac{2}{3} b^{n(n-3)} a^{2n} (n^s abck) + \frac{5}{6} a^{n(n-1)} (n^s abck) \equiv 0 \pmod{n^{s+1}} \\
\Rightarrow & H(c^n, b) + \frac{2a^{2n} [a^{n(n-4)} c^n - b^{n(n-4)} b^n]}{3} (n^s abck) + \frac{5}{6} a^{n(n-1)} (n^s abck) \equiv 0 \pmod{n^{s+1}} \\
\Rightarrow & H(c^n, b) + \frac{2a^{n(n-2)} (c^n - b^n)}{3} (n^s abck) + \frac{5}{6} a^{n(n-1)} (n^s abck) \equiv 0 \pmod{n^{s+1}} \\
\Rightarrow & H(c^n, b) + \frac{2a^{n(n-2)} a^n}{3} (n^s abck) + \frac{5}{6} a^{n(n-1)} (n^s abck) \equiv 0 \pmod{n^{s+1}} \\
\Rightarrow & H(c^n, b) + \frac{3}{2} a^{n(n-1)} (n^s abck) \equiv 0 \pmod{n^{s+1}} \quad (23)
\end{aligned}$$

$$\text{Vì } a, b \text{ đối xứng nhau, nên ta cũng có } H(c^n, a) + \frac{3}{2} b^{n(n-1)} (n^s abck) \equiv 0 \pmod{n^{s+1}} \quad (24)$$

Trừ vế theo vế hai PT đồng dư (23) và (24), ta có:

$$\begin{aligned}
& [H(c^n, b) - H(c^n, a)] + \frac{3}{2} [a^{n(n-1)} - b^{n(n-1)}] (n^s abck) \equiv 0 \pmod{n^{s+1}} \\
\Rightarrow & H(c^n, b) - H(c^n, a) \equiv 0 \pmod{n^{s+1}} \quad (25)
\end{aligned}$$

Theo tính chất của $H(h, b)$ và $H(h, a)$ ứng với $h = c^n$ (PT (17) và PT(17')) và PT (25) ta có

$$\begin{aligned}
& \frac{(1-n)a^n [a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)}]}{(c^n - b^n)^3} - \frac{(1-n)b^n [a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)}]}{(c^n - a^n)^3} \equiv 0 \pmod{n^{s+1}} \\
\Rightarrow & [a^n (c^n - a^n)^3 - b^n (c^n - b^n)^3] [a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)}] \equiv 0 \pmod{n^{s+1}}
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow [a^n(b^n + 2n^s abck)^3 - b^n(a^n + 2n^s abck)^3][a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)}] \equiv 0 \pmod{n^{s+1}} \\
&\Rightarrow (a^n b^{3n} - b^n a^{3n})[a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)}] \equiv 0 \pmod{n^{s+1}} \text{ (vì } a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv 0 \pmod{n^2} \text{), bổ đề 7)} \\
&\Rightarrow -a^n b^n (a^n - b^n)(a^n + b^n)[a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)}] \equiv 0 \pmod{n^{s+1}} \\
&\Rightarrow a^{n(n-2)} + b^{n(n-2)} - c^{n(n-2)} \equiv 0 \pmod{n^{s+1}} \text{ (27) (vì } a^n b^n (a^n - b^n)(a^n + b^n) \not\equiv 0 \pmod{n} \text{)} \\
&\Rightarrow H(c^n, b) \equiv H(c^n, a) \equiv 0 \pmod{n^{s+1}} \text{ (suy ra từ (26) và (27))} \\
&\Rightarrow \frac{3}{2} a^{n(n-1)} (n^s abck) \equiv 0 \pmod{n^{s+1}} \text{ (kết hợp với (25))}
\end{aligned}$$

$\Rightarrow n = 3$, (vô lý)

1.3. Nếu $h = n^{ns-1}c^n$

Ta có

$$\begin{aligned}
* a^{n(n-2)} + b^{n(n-2)} &= [(a^n + b^n) - b^n]^{n-2} + b^{n(n-2)} = \sum_{i=0}^{n-2} (-1)^i C_{n-2}^i (a^n + b^n)^{n-2-i} b^{ni} + b^{n(n-2)} = \\
&= \sum_{i=0}^{n-4} (-1)^i C_{n-2}^i (n^{ns-1}c^n - 2n^s abck)^{n-2-i} b^{ni} + (n-2)(n^{ns-1}c^n - 2n^s abck)b^{n(n-3)} \equiv 0 \pmod{n^{2s}} \\
&\Rightarrow a^{n(n-2)} + b^{n(n-2)} \equiv (n-2)(n^{ns-1}c^n - 2n^s abck)b^{n(n-3)} \equiv 4b^{n(n-3)}n^s abck \pmod{n^{s+1}}, \text{ suy ra } a^{n(n-2)} + b^{n(n-2)} \equiv 0 \pmod{n^s} \text{ (28)} \\
\text{và } a^{n(n-2)} + b^{n(n-2)} &\equiv 4b^{n(n-3)}n^s abck \pmod{n^{s+1}} \text{ (29)}
\end{aligned}$$

$$\begin{aligned}
* a^{n(n-4)} + b^{n(n-4)} &= [(a^n + b^n) - b^n]^{n-4} + b^{n(n-4)} = \sum_{i=0}^{n-4} (-1)^i C_{n-4}^i (a^n + b^n)^{n-4-i} b^{ni} + b^{n(n-4)} = \\
&= \sum_{i=0}^{n-5} (-1)^i C_{n-4}^i (n^{ns-1}c^n - 2n^s abck)^{n-4-i} b^{ni} \equiv 0 \pmod{n^s} \Rightarrow a^{n(n-4)} + b^{n(n-4)} \equiv 0 \pmod{n^s} \text{ (30)}
\end{aligned}$$

$$\begin{aligned}
* a^{n(n-3)} - b^{n(n-3)} &= [(a^n + b^n) - b^n]^{n-3} - b^{n(n-3)} = \sum_{i=0}^{n-3} (-1)^i C_{n-3}^i (a^n + b^n)^{n-3-i} b^{ni} - b^{n(n-3)} = \\
&= \sum_{i=0}^{n-4} (-1)^i C_{n-3}^i (n^{ns-1}c^n - 2n^s abck)^{n-3-i} b^{ni} \equiv 0 \pmod{n^s} \Rightarrow a^{n(n-3)} \equiv b^{n(n-3)} \pmod{n^s} \text{ (31)}.
\end{aligned}$$

$$\begin{aligned}
* a^{n(n-5)} - b^{n(n-5)} &= [(a^n + b^n) - b^n]^{n-5} - b^{n(n-5)} = \sum_{i=0}^{n-5} (-1)^i C_{n-5}^i (a^n + b^n)^{n-5-i} b^{ni} - b^{n(n-5)} = \\
&= \sum_{i=0}^{n-6} (-1)^i C_{n-5}^i (n^{ns-1}c^n - 2n^s abck)^{n-5-i} b^{ni} \equiv 0 \pmod{n^s} \Rightarrow a^{n(n-5)} \equiv b^{n(n-5)} \pmod{n^s} \text{ (32)}
\end{aligned}$$

Từ (16), (28), (29), (30), (31), (32), $h - b^n \equiv a^n \equiv -b^n \pmod{n^s}$ và $h = n^{ns-1}c^n \equiv 0 \pmod{n^{ns-1}}$, Từ (16'') suy ra

$$\begin{aligned}
&\frac{(h-b^n)H(h,b)}{(h-b^n)} - \frac{15b^{n(n-3)}(h-b^n)^2}{(h-b^n)^4}n^s abck + \frac{5b^{n(n-2)}(h-b^n)}{3(h-b^n)^4}(n^s abck) - \frac{2b^{n(n-3)}(h-b^n)^2 - 18b^{n(n-3)}(h-b^n)^2}{3(h-b^n)^4}(n^s abck) + \\
&+ \frac{4(-b^{n(n-2)})(h-b^n)(n^s abck)}{(h-b^n)^4} + \frac{(-b^{n(n-4)})(h-b^n)^3}{4(h-b^n)^4}(n^s abck) + \frac{9b^{n(n-4)}(-b^n)(h-b^n)^2 - 3[2b^{n(n-3)}(h-b^n)^2]}{2(h-b^n)^4}(n^s abck) + \\
&+ \frac{-7b^{n(n-4)}(h-b^n)^3 + 7b^{n(n-3)}(h-b^n)^2 - 7[b^{n(n-2)}(h-b^n) - b^{n(n-1)}]}{(h-b^n)^4}(n^s abck) + \frac{4b^{n(n-3)}(h-b^n)^2}{3(h-b^n)^4}(n^s abck) + \\
&+ \frac{2b^{n(n-3)}(h-b^n)^2(n^s abck)}{(h-b^n)^4} + \frac{7b^{n(n-2)}(h-b^n)}{(h-b^n)^4}(n^s abck) + \frac{5[3b^{n(n-4)}(h-b^n)^3 - 4b^{n(n-3)}(h-b^n)^2 + 6b^{n(n-2)}(h-b^n)]}{4(h-b^n)^4}(n^s abck) + \\
&+ \frac{b^{n(n-5)}(h-b^n)^4}{4(h-b^n)^4}(n^s abck) + \frac{8[b^{n(n-3)} + b^{n(n-3)}](n^s abck)b^{2n}}{(h-b^n)^4} \equiv 0 \pmod{n^{s+1}} \text{ (17'')}
\end{aligned}$$

$$\begin{aligned}
\text{Ta có } H(h, b) &\equiv \frac{(1-n)a^n[a^{n(n-2)} + b^{n(n-2)} - h^{n-2}]}{(h-b^n)^3} \pmod{n^{s+1}} \text{ (tính chất của } H(h, b)) \\
\Rightarrow \frac{(h-b^n)H(h, b)}{h-b^n} &\equiv \frac{(1-n)(-b^n)[a^{n(n-2)} + b^{n(n-2)}](-b^n)}{(h-b^n)^4} \equiv \frac{b^{2n}(4b^{n(n-3)}n^s abck)}{(h-b^n)^4} \pmod{n^{s+1}} \\
\Rightarrow \frac{(h-b^n)H(h, b)}{h-b^n} &\equiv \frac{4b^{n(n-1)}n^s abck}{(h-b^n)^4} \pmod{n^{s+1}} \text{ (17''') Kết hợp (17'') và (17''')} \text{ ta có:} \\
&\frac{4b^{n(n-1)}n^s abck}{(h-b^n)^4} + \frac{5[3b^{n(n-4)} \cdot (-b^{3n}) - 4b^{n(n-3)} \cdot b^{2n} + 6b^{n(n-2)} \cdot (-b^n)]}{4(h-b^n)^4} (n^s abck) + \frac{4b^{n(n-3)} \cdot b^{2n}}{3(h-b^n)^4} (n^s abck) - \frac{15b^{n(n-3)} \cdot b^{2n}}{(h-b^n)^4} n^s abck + \\
&+ \frac{4(-b^{n(n-2)})(-b^n)(n^s abck)}{(h-b^n)^4} + \frac{(-b^{n(n-4)})(-b^n)^3}{4(h-b^n)^4} (n^s abck) + \frac{9b^{n(n-4)}(-b^n)b^{2n} - 6b^{n(n-3)} \cdot b^{2n}}{2(h-b^n)^4} (n^s abck) + \frac{b^{n(n-5)}b^{4n}}{4(h-b^n)^4} (n^s abck) + \\
&+ \frac{-7b^{n(n-4)}(-b^n)^3 + 7b^{n(n-3)} \cdot b^{2n} - 7[b^{n(n-2)}(-b^n) - b^{n(n-1)}]}{(h-b^n)^4} (n^s abck) + \frac{7b^{n(n-2)}(-b^n)}{(h-b^n)^4} (n^s abck) + \frac{5b^{n(n-2)}(-b^n)}{3(h-b^n)^4} (n^s abck) \\
&+ \frac{2b^{n(n-3)} \cdot b^{2n} (n^s abck)}{(h-b^n)^4} - \frac{2b^{n(n-3)} \cdot b^{2n} - 18b^{n(n-3)}b^{2n}}{3(h-b^n)^4} (n^s abck) + \frac{8[b^{n(n-3)} + b^{n(n-3)}](n^s abck)b^{2n}}{(h-b^n)^4} \equiv 0 \pmod{n^{s+1}} \\
\Rightarrow 4b^{n(n-1)} - \frac{65b^{n(n-1)}}{4} + \frac{4b^{n(n-1)}}{3} + 4b^{n(n-1)} + \frac{b^{n(n-1)}}{4} - \frac{15b^{n(n-1)}}{2} + 28b^{n(n-1)} - 15b^{n(n-1)} + \frac{b^{n(n-1)}}{4} + 2b^{n(n-1)} + \frac{16b^{n(n-1)}}{3} - \\
- 7b^{n(n-1)} - \frac{5b^{n(n-1)}}{3} + 16b^{n(n-1)} \equiv 0 \pmod{n} \\
\Rightarrow \frac{55}{3}b^{n(n-1)} \equiv 0 \pmod{n}
\end{aligned}$$

$\Rightarrow n = 11$ hoặc $n = 5$ (vô lý).

Vậy, trong trường hợp $k \neq 0$, PT (1) vô nghiệm nguyên khác không, khi n là số nguyên tố lớn hơn 11.

Tóm lại, PT (1) vô nghiệm nguyên khác không, khi n là số nguyên tố lớn hơn 11.

2. Khi n là số nguyên tố nhỏ hơn 13

Nhà toán học **Pierre de Fermat (1601 – 1665)** đã chứng minh PT(1) vô nghiệm nguyên khác không, khi $n = 4$.

Nhà toán học **Leonhard Euler (1707 – 1783)** đã chứng minh PT(1) vô nghiệm nguyên khác không, khi $n = 3$.

Nhà toán học **Dirichlet (1805 – 1859)** đã chứng minh PT(1) vô nghiệm nguyên khác không, khi $n = 5$.

Nhà toán học **Gabriel Lamé (1795 – 1870)** đã chứng minh PT(1) vô nghiệm nguyên khác không, khi $n = 7$.

Nhà toán học người **Ernst Kummer (1810 – 1893)** đã chứng minh PT(1) vô nghiệm nguyên khác không, khi $n = 11$.

3. Khi $n = pk$, p là số nguyên tố lẻ hoặc $p = 4$ ($k \in \mathbb{N}^*$)

Từ kết quả chứng minh ở 1., 2., áp dụng **bổ đề 2**, ta có PT (1) vô nghiệm nguyên khác không.

Vậy định lý lớn Fermat đã được chứng minh.

Phú Vang, ngày 22/9/2026

Tác giả: **Tôn Thất Hiệp**